15 Mission Analysis

Overview

Mission analysis refers to the complete set of assumptions regarding the operation of an aircraft and the resulting weight and fuel planning. Several parametric considerations can be added, including mission optimisation, environmental emissions and various commercial or military trade-offs. This chapter illustrates how a detailed mission analysis can be carried out. We start by defining mission scenarios (§ 15.1) and the payload-range charts (§ 15.2). We then make a detailed qualitative and quantitative mission analysis (§ 15.3), including the problems of mission range and mission fuel (§ 15.4), the fuel-reserve policies (§ 15.5) and the short-range performance at MTOW (§ 15.6). In § 15.7 we discuss more advanced problems, such as service with an intermediate stop (§ 15.7.1), tankering (§ 15.7.2) and the point-of-no-return (§ 15.7.3). We proceed with the calculation of the direct operating costs (§ 15.8). We show some case studies: an aircraft selection problem (§ 15.9), a fuel mission planning for a transport aircraft (§ 15.10) and the effect of floats on the payload-range chart of a propeller airplane adapted with floats for water operations (§ 15.11). We close this chapter with an introduction to risk analysis in aircraft performance (§ 15.12).

KEY CONCEPTS: Mission Profiles, Flight Scheduling, Payload-Range Charts, Mission Analysis, Reserve Fuel, Fuel Planning, Tankering, En-Route Stop, Direct Operating Costs, Risk Analysis, Route Selection.

15.1 Mission Profiles

A mission profile is a scenario that is required to establish the weight, fuel, payload, range, speed, flight altitude, loiter and any other operations that the aircraft must be able to accomplish. The mission requirements are specific to the type of aircraft. For high-performance aircraft they get fairly complicated and require some statistical forecasting. The deterministic approach is not sufficient or even recommended.

Over the years many commercial aircraft operators have specialised in niche markets that offer prices and services to selected customers. These niches include the executive jet operators between major business centres, operators flying to particular destinations (oil and gas fields), the all-inclusive tour operators to holiday resorts

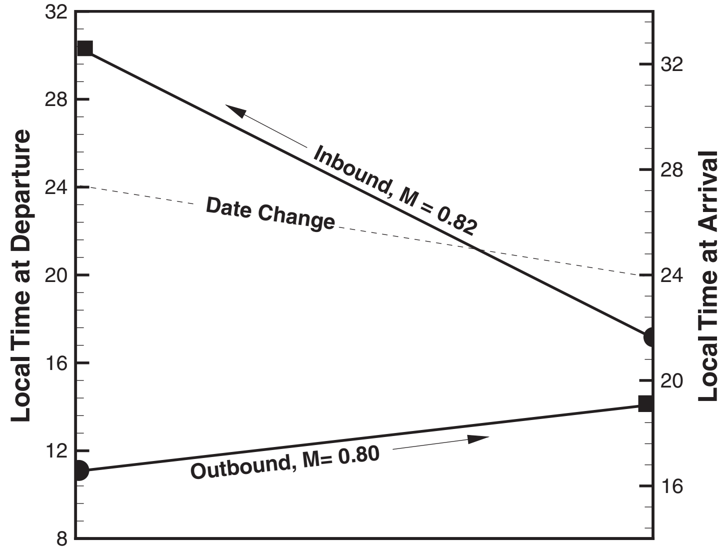

Figure 15.1. Scheduling of trans-Atlantic flight.

and the no-frills airlines, flying to minor and under-used airports. These operators have different schedules and cost structures.

Let us start with long-range passenger operations, which are serviced by subsonic commercial jets. The basic principle is that the airplane takes off from airport A and flies to airport B along a recognised flight corridor; then it returns to A. The main parameters of the mission planning are the distance between the airports, the flight time, the down time at the airport for getting the airplane ready (also called *time-on-station*), the flight speed, the local air traffic and the departure times at both ends. Back at the airport of origin, the day is not over; the operator may wish to utilise the airplane for another flight to the same destination or to another destination. The key is the departure time and the minimisation of the curfew. Figure 15.1 shows a typical schedule of such an airplane over a trans-Atlantic route from a major airport in Europe to an airport on the East Coast of the United States.

Due to the time-zone effect, a late-morning departure from Europe arrives to the United States in the mid-afternoon. An early-evening departure arrives back in Europe in the early hours of the day after. Over a 24-hour period the airplane will have done a return flight and worked about 14 to 16 hours. For a flight arriving late in the evening, a return may not be possible until early morning on the next day. This adds to the operational costs because of the need of maintaining the crew away from the home port.

The schedule shown in Figure 15.2 shows a typical schedule between a UK airport and mainland Europe. Due to the relatively short distance of 600 n-miles and one time zone, it is possible to operate two return services and one outbound service, whilst maintaining the night-time curfew (shaded area). The final outbound flight must remain at the destination airport till the following morning. This aircraft will be the first to make an inbound flight. To run efficiently, this service would require two aircraft, one at each airport. If this is not possible, then the last outbound flight must be cancelled from the schedule.

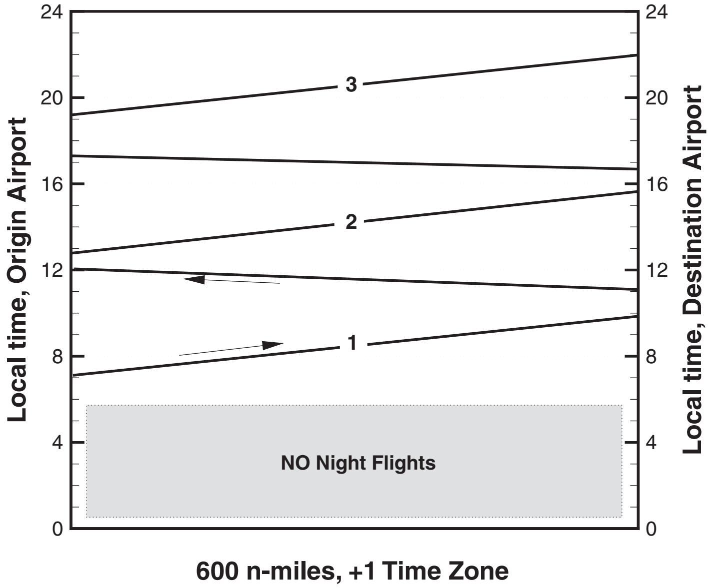

Figure 15.2. Scheduling of short–medium flight.

Another key factor in the utilisation of the airplane is the turn-around time, that is, the time required to prepare the airplane for the next flight. Considerable logistics are required to ensure all services to passengers, cargo and airplane within a target time frame. Up to six different categories of services are involved: passenger boarding/disembarking, cargo loading/unloading, refuelling, cleaning, catering, and ground handling. The time needed to get the aircraft ready for the next intercontinental flight may require up to 3 hours. However, Airbus reports that the turn-around time of the A380 is 90 minutes with standard servicing on two decks and 120 minutes with servicing from the main deck*.

15.1.1 Operational Parameters

Block time is the time required to travel from the departure gate to the destination gate. If the block time of each flight is added, at the end of one year the airplane will have a total number of block hours, which airline operators may want to maximise. There are three parts in the block time: taxi out, airborne time and taxi in. An airplane flying a day-time shorter route, and returning in the mid-afternoon, should be able to complete another return flight to the same destination or otherwise. For an airline company operating several airplanes, scheduling and optimal operation of the fleet is a complex problem. Events such as bad weather can lead to dozens of airplanes and flight crews out of position for several days. Scheduling and operation of aircraft is a subject for operations research and is addressed by specialised publications, for example Gang Yu[1], who deals with demand forecasting, network design, route planning, airline schedule planning, irregular operations, integrated scheduling, airport traffic simulation and control, and more. Coy[2] presents a method for the statistical

* Airbus, *A380 Airplane Characteristics*, Rev. Oct. 01/09 (2009), Blagnac Cedex, France.

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prediction of the block time, which, unlike the deterministic scenario shown in Figure 15.1, relies on a number of stochastic parameters, including congestion, weather conditions and delays accumulated on an earlier leg.

In the following discussion, we present only a deterministic scenario and focus on performance*. For a transport-type airplane, there are two basic types of mission analysis:

- **Range Planning**: For given atmospheric conditions, given the payload and the fuel load, calculate the distance that can be flown. The problem is to establish the functional relationship between the flight plan and the operational parameters:

$$X = f(W_p, W_f, \mathtt{Atm}). \tag{15.1}$$

- **Fuel Planning**: For given atmospheric conditions, given the required distance X and the payload weight W_p, calculate the mission fuel W_f and the weight breakdown. The problem is to establish the functional relationship between the fuel planning and the key operational parameters:

$$W_f = f(X, W_p, \mathtt{Atm}). \tag{15.2}$$

The operator `Atm` is a vector containing all of the atmospheric conditions encountered during the flight. For convenience, these conditions are split between 1) take-off and climb-out, 2) cruise, and 3) final approach and landing. More specifically, the parameters are atmospheric temperatures, wind speed, wind direction and relative humidity. The latter parameter is only relevant for noise calculations (see Chapters 16 to 18).

15.2 Range-Payload Chart

The payload-range chart belongs to the range flight-performance analysis (Equation 15.1) because the distance that can be flown is unknown. The flight distance depends not only on the amount of fuel but also on the weight of the payload. In most cases the combination of maximum payload and maximum fuel load exceeds the MTOW. If we know how to calculate the flight range, then it is possible to construct a chart showing how some of the aircraft weights are related to the range. Because different cruise techniques are available, we could construct different weight-payload diagrams corresponding to each flight program. Furthermore, there are effects of atmospheric conditions, climb and descent techniques, and reserve fuel policy that can change the weight-payload performance considerably. Unless all of these conditions are specified, it is not possible to compare the range-payload performance of two aircraft.

There are several ways in which the payload-range performance can be analysed. The first approach is shown in Figure 15.3, which displays the range-payload chart for three commercial subsonic jet aircraft of the Airbus family. The manufacturer generally specifies the range at maximum passenger load (including baggage) and some value of the bulk payload; the latter item is important for most commercial airlines. However, passenger-carrying airplanes seldom operate at MTOW.

* We refer to "mass" and "weight" interchangeably, although they are physically different. The aviation jargon refers to weights rather than mass; our computer models work with "mass".

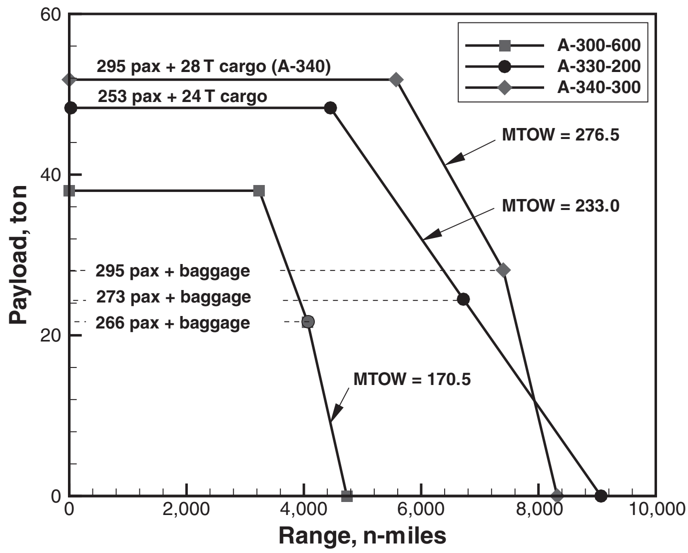

Figure 15.3. Maximum payload range for Airbus airplanes.

Another way of showing the payload-range performance is presented in Figure 15.4. The data have been elaborated from Boeing[3] and refer to the B747-400. This diagram is considerably different and more detailed than Figure 15.3. The oblique lines denote constant-BRGW values. A trade between an increase in range and the corresponding decrease in payload shall be made at constant brake-release gross weight:

$$\left(\frac{d\,\mathrm{ZFW}}{dX}\right)_{BRGW}.$$

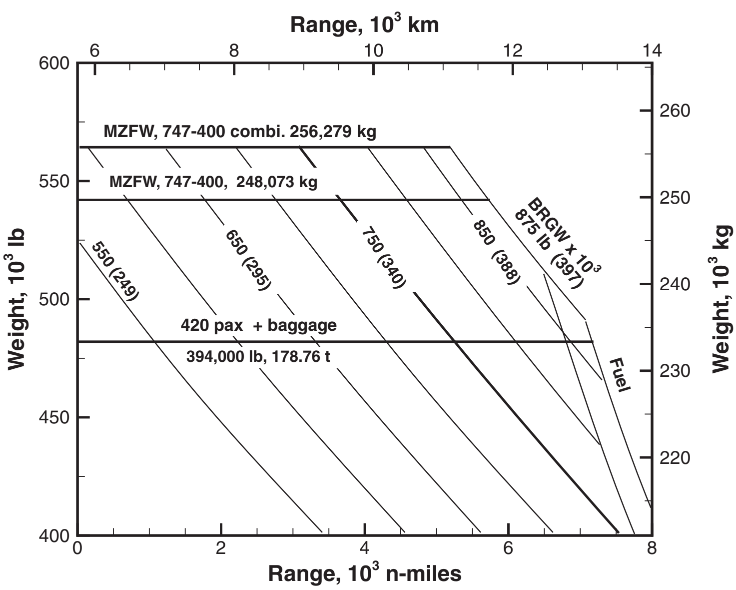

Figure 15.4. Payload-range chart for the Boeing B-747-400 and -400 Combi (CF6-80C2B1F engines); standard day, $M = 0.85$, cruise-climb profile; FAR international reserves.

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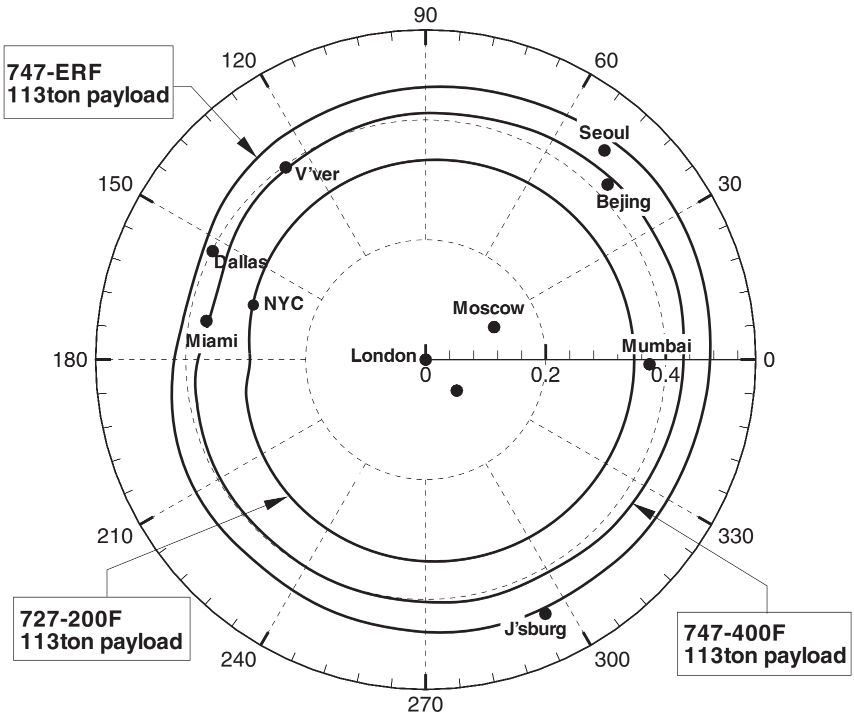

Figure 15.5. Range of Boeing 747-family of aircraft. Data elaborated from Boeing.

The ZFW is the weight of the aircraft except the loaded fuel; it includes the bulk payload, the crew and all of the operational items. At a constant BRGW, the range must decrease and, therefore, this derivative must be negative; its value is the *sensitivity* of the range to changes in ZFW at constant BRGW.

The key parametric on the flight-range effect is the wind. Figure 15.5 shows some destinations that can be reached with the Boeing 747. The data have been extrapolated from the manufacturer and include typical mission rules, 85% annual winds and air-traffic allowances. This type of chart is more useful from a marketing point of view than an engineering point of view because it gives very few details on the bulk payload, the passenger load, the flight procedures, the availability of flight corridors and so on.

Another example of a payload-range chart is shown in Figure 15.6 that refers to the business jet airplane Gulfstream G550†. The data are presented in the range-GTOW plane at selected values of the payload. The flight time corresponding to a flight distance is useful in general flight planning (such as in Figure 15.1). The fuel load has to be inferred from the data. For example, for a specified payload W_p and required range X, the diagram returns a weight $W =$ GTOW. The corresponding fuel weight W_f is found from the difference:

$$W_f = W - W_e - W_p. \tag{15.3}$$

The operating empty weight W_e is given in the airplane documents (FCOM, type certificate).

† Gulfstream G550. Flight Crew Operating Manual (2005).

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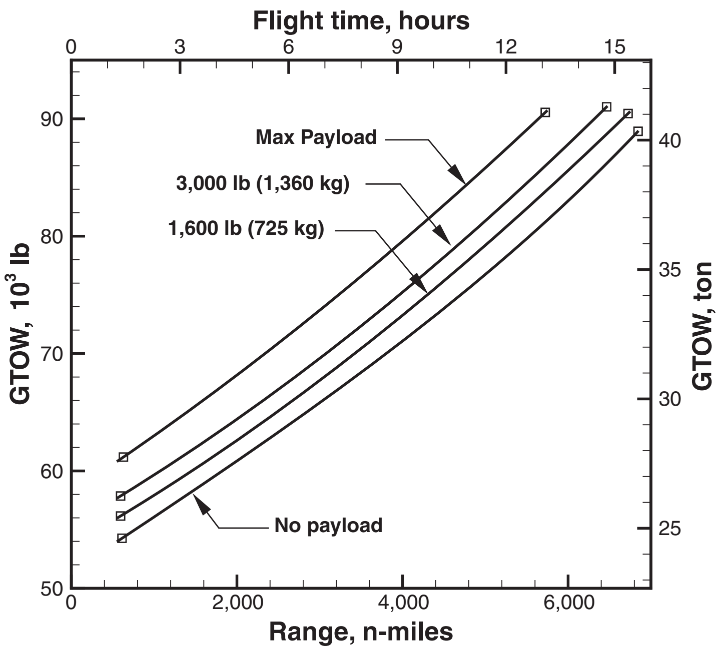

Figure 15.6. Payload-range of Gulfstream G550. Data elaborated from Gulfstream Aerospace; $M = 0.80$; 200 n-miles diversion fuel, standard day, no wind.

From Equation 15.3 the fuel fraction $\xi = W_f/W$ (fuel weight over gross weight) can be written as

$$\xi = 1 - \frac{W_e}{W} - \frac{W_p}{W}. \tag{15.4}$$

Because the empty weight is fixed, Equation 15.4 is a linear relationship between payload fraction and fuel fraction, with the gross weight W being a parameter.

15.2.1 Case Study: Range Sensitivity Analysis

For a flight range of 5,000 n-miles (point X in Figure 15.7), the difference in GTOW between an airplane loaded to its maximum payload weight (point A) and an airplane loaded with 1,360 kg (3,000 lb) of payload (point B) is 2,052 kg (4,520 lb). The certified maximum payload is 2,812 kg (6,200 lb). This means that the difference in payload between the two cases is 760 kg (1,680 lb):

$$\Delta \text{GTOW} = 2{,}052\,\text{kg}, \qquad \Delta W_p = 760\,\text{kg}, \qquad \Delta W_f = 2{,}052 - 760 = 1{,}292\,\text{kg}.$$

This means that an increase of the payload by 760 kg requires an additional ~1,300 kg of fuel to cover the same distance, or

$$\left(\frac{dW_f}{dW_p}\right)_X = 1.7. \tag{15.5}$$

This is by any account a modest value, especially when compared to the overall operating costs of the airplane. Alternatively, consider the case of two flights at the same payload (points B and C in Figure 15.7).

$$\Delta X = 719\,\text{n-miles}, \qquad \Delta \text{GTOW} = 0, \qquad \Delta W_f = -\Delta W_p = 760\,\text{kg}.$$

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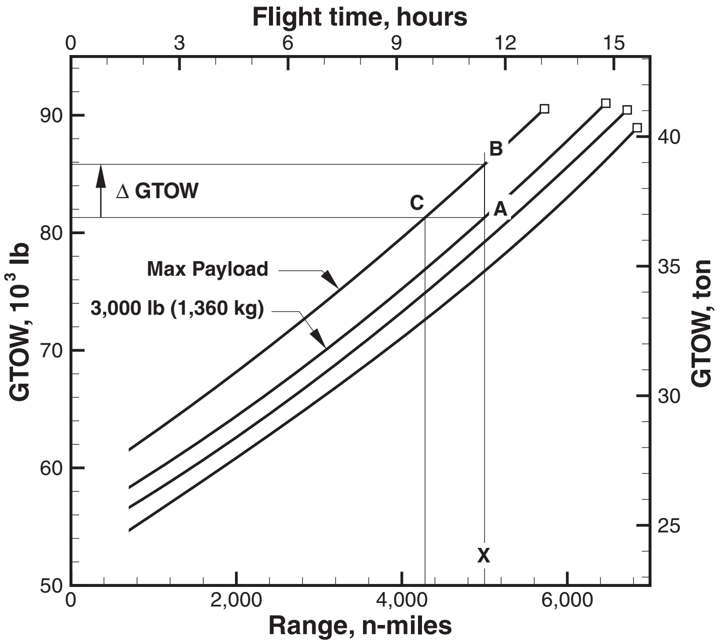

Figure 15.7. Analysis of payload-range sensitivity of Gulfstream G550.

Thus, the decrease in payload is offset by an increase in fuel, which leads to the following equivalences:

$$d\,\mathrm{ZFW} = d\,(W_e + W_p) = dW_p = -dW_f, \tag{15.6}$$

$$\left(\frac{d\,\mathrm{ZFW}}{dX}\right)_{GTOW} = -\frac{dW_f}{dX} = -1.057\,\mathrm{kg/n\text{-}mile}. \tag{15.7}$$

15.2.2 Case Study: Payload-Range of the ATR72-500

We used the ATR72-500 powered by PW127M turboprop engines and Hamilton F568-1 propellers. This configuration has been discussed in other sections of the book. We calculated the payload-range performance for two different weights and compared our results to the "official" performance quoted by the manufacturer. These results are shown in Figure 15.8. The calculation consists of four key points: TOW-limited range, maximum-payload range, maximum-fuel range and ferry range.

15.2.3 Calculation of the Payload-Range Chart

We now proceed to a practical method for the calculation of the payload-range chart. To avoid computer-intensive calculations, some simplifications are required. We need to define three critical operational points: 1) the maximum-payload range X_{p1}; 2) the maximum-fuel range X_{p2}; and 3) the ferry range X_{p3}. The operation at ranges between X_{p1} and X_{p2} requires to consider the aircraft at constant BRGW. The difference is only marginal, but the additional computational burden would be considerable.

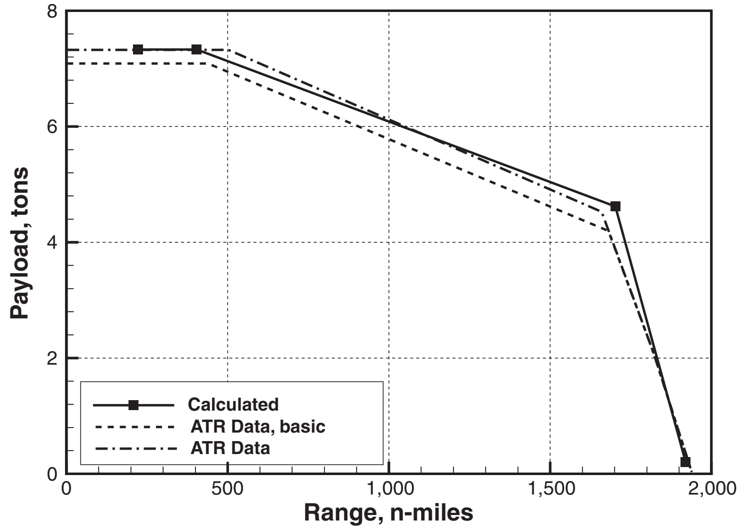

Figure 15.8. Payload-range performance of the ATR72-500.

1. **Maximum-Payload Range:** Assume that there are no passengers; assume a bulk payload equal to the MSP. Then use a bisection method to calculate the fuel planning for a tentative range X with the constraint GTOW = MTOW. In other words, if the fuel planning corresponding to the tentative range X leads to a GTOW < MTOW, then increase the range; otherwise decrease the range. The maximum-payload range is bracketed by $X_1 = X_{design}/2$ and $X_2 = X_{design}$. This procedure converges to X_{p1} in 7 to 10 iterations, not without difficulties (it is possible that the aircraft is unable to climb to the estimated ICA, or that the GTOW corresponding to the range X is considerably larger than the MTOW).
2. **Maximum-Fuel Range:** Assume GTOW = MTOW. In this case, as the range increases, payload has to be traded for fuel. If W_{mfw} is maximum fuel capacity, the bulk payload must be

$$W_p = \text{GTOW} - W_e - W_{mfw}.$$

It is not useful to add passenger weight at this time because passengers require consideration of on-board services. This is a fixed quantity for the maximum-fuel range X_{p2}. After bracketing the range $X_{p1} < X_{p2} < X_{design}$, use the bisection method to calculate the maximum-fuel range. For a given iteration, if GTOW < MTOW, increase the tentative range, otherwise decrease the range. The procedure should converge to X_{p2} in less than 10 iterations unless numerical difficulties intervene (such as at the previous point).
3. **Ferry Range:** There is no payload in this case. Thus, the take-off weight must be

$$\text{GTOW} = W_e + W_{mfw}.$$

After bracketing the solution with $X_{p2} < X_{p3} < 1.1X_{design}$, use the bisection method to calculate ferry range. For a tentative range if the mission fuel is lower than the fuel capacity, the range must increase; otherwise it must decrease.
4. **Maximum-Passenger Range:** Along with the three points specified previously, a passenger aircraft must show the design range at maximum passenger load.

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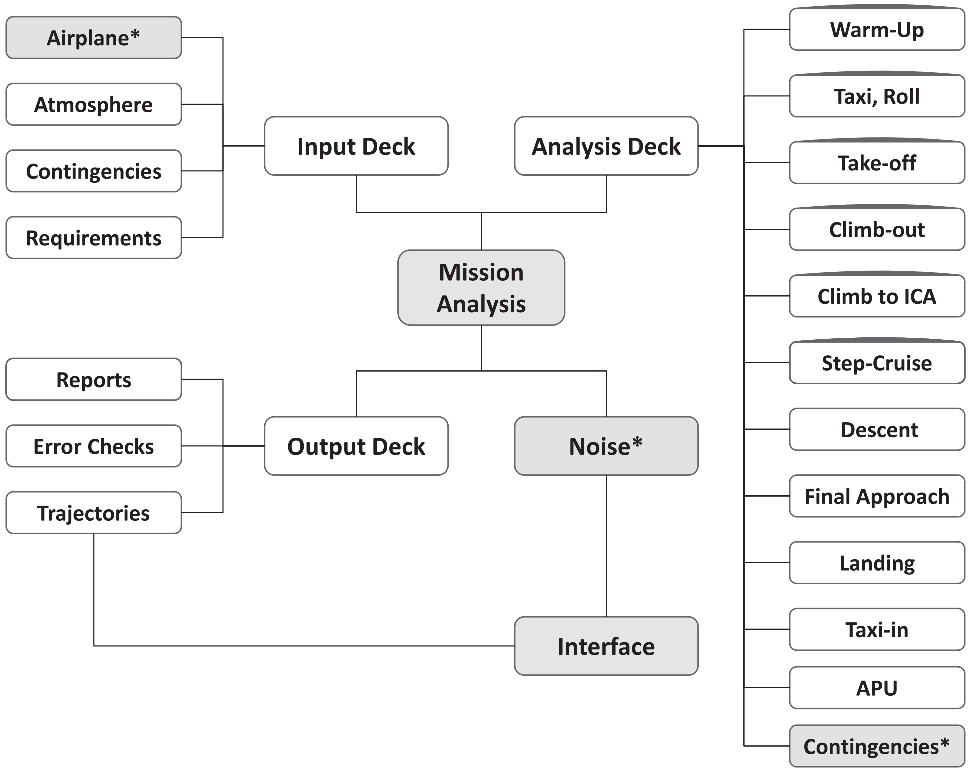

Figure 15.9. Flowchart of a mission-fuel analysis.

This range is usually between X_{p1} and X_{p2}. For this point, the payload weight is the sum of all passengers, their baggage, average weight of service items and average number of crew members (and associated weight). After bracketing the solution with $X_{p1} < X < X_{p2}$, the solution procedure follows the same steps as in the previous scenarios.

The calculation of a payload-range must rely on a number of assumptions, some of which can be arbitrary. The fuel reserve (§15.5) and the atmospheric conditions must be specified.

15.3 Mission Analysis

A realistic flight planning requires an optimisation of the route (track, vertical profile), Mach number and altitude, with the constraints imposed from the air traffic control and the aviation regulators. The flight planning must be based on real-time data, such as winds, temperatures, payload and other parameters. Therefore, a detailed analysis involves several parameters. A flowchart of the computer program is shown in Figure 15.9.

The boxes with an asterisk refer to the sub-models: the airplane model is described in Chapter 2; the aircraft noise model is discussed in Chapters 16 and 17 (see also flowchart in Figure 16.3). The fuel contingencies are discussed in § 15.10. An interface is required to associate a flight trajectory to a noise calculation (§ 18.3).

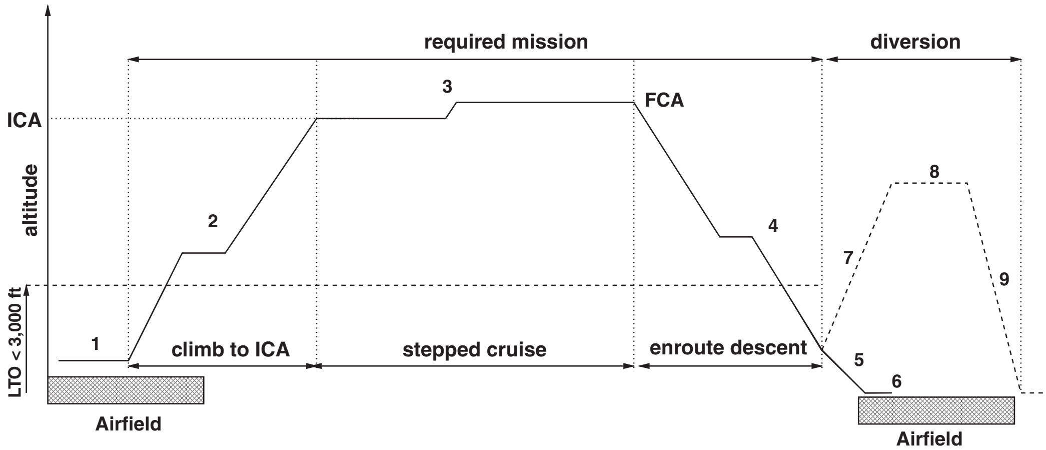

Figure 15.10. Standard mission profile of passenger aircraft, from Ref.[4].

A typical mission profile for a commercial jet is shown in Figure 15.10. It shows a climb to cruise altitude, a cruise and a descent. It also includes the case of an aborted landing and an extension of the flight at a lower altitude. Diagrams of this nature can be made more precise by considering a quantitative scale for the distance and altitude, and for each flight segment. More specifically, there are two sub-problems:

- Mission range for given fuel and payload without constraint on BRGW, § 15.3.1.
- Mission fuel for given range and payload with constraint on BRGW, § 15.4.

The weight constraint can be specified on the BRGW or the GTOW. The second case is only constrained by the fact that GTOW $\leq$ MTOW.

Calculation of the exact fuel required for a mission requires consideration of several objectives, some of which can be conflicting. Safety comes first. After that, the airline must consider its profitability via its own accounting system and punctuality of the service. Other items include actual payload, weather phenomena, operational constraints, route and altitude constraints. The cost of fuel is in most cases the largest cost item of a commercial airline. Price volatility makes planning difficult and risky. Fuel planning itself requires three phases:

1. Pre-flight: It refers to measures in terms of route planning, ground operations, usage of APU versus ground services, turn-around processes and other administrative tasks.
2. Flight: It refers to all of the flight procedures to be programmed in the flight computer, such as planned route, climb and descent schedules, flight level selection and so on.
3. Post-flight: It includes the collection and analysis of flight data that are used to implement strategies for reduction of fuel consumption. These measures include monitoring of the engines, the airframe and the systems from the FDR.

Calculation of the mission fuel (point 2) requires adding up the fuel required by each segment of the flight. For a passenger operation, the mission requirement

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is straightforward. The fuel for the APU must be considered as well because for a large aircraft, this means a considerable amount of extra weight. The APU fuel flow can be of the order of 150 kg/h; for a 7-hour intercontinental flight of a wide-body commercial airliner, this means 1,050 kg, or the weight of 12 paying passengers.

Our numerical procedure is based on a combination of optimal fuel planning and optimal flight trajectories. Each segment is treated with some degree of independence. For example, the initial cruise condition is established on the basis of the optimal SAR with the estimated AUW at the top of climb. Likewise, the descent trajectory is based on initial conditions established from the cruise. Alternative methods have been proposed, including the use of multi-variate gradient optimisation[5]. These methods, aided by the use of constraints, offer some advantages in global optimisation.

15.3.1 Mission Range for Given Fuel and Payload

The problem is to establish how far the aircraft can fly, with the mandatory fuel reserves and under specified atmospheric conditions. The numerical procedure is more elaborate than the calculation of a cruise range because of the implicit relationship between airplane trajectories (climb, cruise, descent) and BRGW. We propose the following algorithm:

1. Calculate the fuel from ramp up to the ICA. This operation includes taxi-out, take-off, climb to ICA and APU fuel to that point. Call this fuel $W_{f_{ICA}}$; x_c is the distance to ICA.
2. Calculate the extended all-out fuel $W_{f_{ext}}$ to take into account holding and diversion procedures. This sub-task is fully specified in § 15.5.
3. Using the procedures indicated in Chapter 11, calculate the descent fuel W_{f_d} and distance x_d from the top of descent (final cruise altitude), with an initial weight

$$W = \text{BRGW} - \underbrace{(W_{f_{ext}} - W_{f_{res}})}_{fuel\ \ remaining}, \tag{15.8}$$

   where $W_{f_{res}}$ is the reserve fuel, which is analysed separately in § 15.5. The weight at the top of descent is equal to the BRGW minus the fuel remaining at the end of the service; this fuel includes the contribution from the diversion and the reserve. If this is the first iteration, start the descent from the ICA (the FCA is indeterminate); otherwise, estimate the FCA from the density altitude. In the latter case, the FCA is found from the condition that W/ρ is a constant, or

$$\sigma_{end} = \sigma_{start} \left(W_{start} - W_{f_{cruise}} \right). \tag{15.9}$$

   The step from σ_{end} to the FCA requires the inverse solution of the atmospheric model. The FCA is then adjusted to the nearest possible flight level (§ 12.7). In any case, add the APU fuel relative to this flight segment.
4. If this is the first iteration, neglect the APU and estimate the fuel available for cruise from

$$W_{f_{cruise}} = W_{f_{usable}} - \left(W_{f_{ICA}} + W_{f_d} + W_{f_{ext}} \right). \tag{15.10}$$

In any other case, add the APU fuel for the duration of the flight.

5. If this is the first iteration, estimate the cruise range with

$$x_{cruise} \simeq \frac{U}{f_j}\left(\frac{L}{D}\right)\log\left(1+\frac{W_{f_{cruise}}}{W_f - W_{f_c}}\right). \tag{15.11}$$

In any other case, calculate the cruise fuel according to the procedures explained in Chapter 12 (stepped-climb).

6. The estimated mission range is

$$X_1 = x_c + x_d + x_{cruise}. \tag{15.12}$$

7. The calculations are repeated from point 2. The stopping criterion is based on the change of X_1 between iterations; when the updated mission range does not change by any meaningful amount (for example, 1 n-mile), the procedure has converged.

Note that the convergence analysis is done on the mission range. An important sub-case is when

$$W_e + W_f + W_p = \text{MRW}. \tag{15.13}$$

When W_p is maximum and W_f is the maximum allowable fuel weight compatible with the MRW, the corresponding range is the *maximum-payload range*. When the fuel weight is maximum and the payload is the maximum allowable weight compatible with the MRW we have the *maximum-fuel range*. Finally, if $W_p = 0$, the ramp weight is most likely below the MRW. The corresponding range is called *maximum-ferry range*.

15.4 Mission Fuel for Given Range and Payload

The problem is to establish the amount of fuel to load onto the aircraft in order to transport a specified payload over a specified distance X_{req}, under specified (or forecast) atmospheric conditions. At the end of the process, we will have a detailed weight breakdown. There are two important steps: 1) make a first estimate of the mission fuel, and 2) improve the first estimate with a suitable numerical method.

15.4.1 Mission-Fuel Prediction

We define the computational steps required in a procedure called `MissionFuel(..)`, which returns the fuel required for a given range, payload, atmospheric conditions and other external factors. The computational procedure is as follows:

1. Estimate the ramp mass (or weight or the BRGW) by making a guess of the fuel required by the mission, W_f^*. This weight is

$$W \simeq W_e + W_p + W_f^*. \tag{15.14}$$

2. Calculate the fuel from the ramp (or gate) up to the ICA. This operation includes taxi-out, take-off, climb to ICA and APU fuel to that point. Call this fuel $W_{f_{ICA}}$; x_c is the distance to ICA.

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3. Calculate the extended all-out fuel $W_{f_{ext}}$ to take into account holding and diversion procedures. This sub-task is fully specified in § 15.5.
4. Using the procedures indicated in Chapter 11, calculate the descent fuel W_{f_d} and distance x_d from the top of descent. Follow the same procedure described at point 3 on page 434.
5. Calculate the cruise distance from

$$x_{cruise} = X_{req} - x_c - x_d. \tag{15.15}$$

6. Calculate the cruise fuel required to fly the distance x_{cruise} from the conditions at ICA (see point 2). This is done with step climbs, following the procedure discussed in § 12.7, by integration of the SAR. Add the APU fuel for this segment.
7. Calculate the ramp weight. This is derived from the sum of all segment fuels.

With reference to point 1 in this procedure, the initial fuel estimate is either provided externally or is specified by the following equation:

$$W_f \simeq c \left(\frac{X_{req}}{X_{design}} \right) c_{usf}, \tag{15.16}$$

where $c_{usf} < 1$ denotes the amount of usable fuel as a portion of the full tanks; the coefficient $c = 0.90$ to 0.95 is a factor that improves the prediction and is found from extensive numerical analysis.

15.4.2 Mission-Fuel Iterations

There are at least three different methods for calculating iteratively the mission fuel:

a) Method based on updating the initial guess until convergence.
b) Method based on updating the latest guess with an under-relaxation.
c) Method based on a predictor-corrector of the initial guess.

These methods are listed from the less robust to the most robust. In some circumstances all of these methods converge, but there is no guarantee that they will always converge. The difficulty increases with the increasing required range and with the size of the airplane because of the increased non-linearity between the fuel weight and the mission fuel.

A) MISSION-FUEL UPDATE. Compare the weight W_i calculated at iteration i with the earlier estimate W_{i-1}. Now use the updated ramp weight to recalculate the mission fuel. The residual is defined as

$$E = 1 - \frac{W_i}{W_{i+1}}. \tag{15.17}$$

There is no guarantee that this procedure converges or that it converges to the correct weight. It depends both on how good is the first estimate and on the required range; it gets worse as the required range is close to the design range. In the latter case, it is possible than on the second iteration, the ramp weight exceeds the MRW and thence the climb becomes sluggish (not enough excess power is available), the

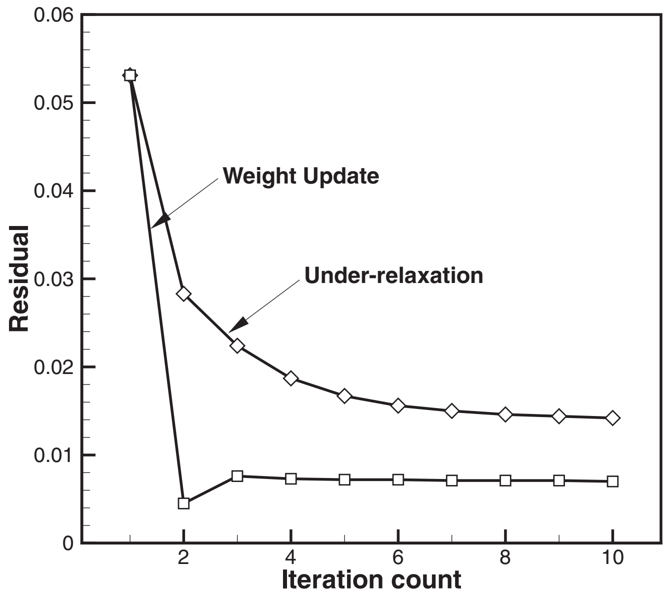


Figure 15.11. Error monitoring, Equation 15.17, in iterative mission-fuel analysis.

best ICA cannot be reached and, ultimately, the computations fail. Therefore, it may be useful to add a constraint so that in no case the ramp weight exceeds the MRW.

B) MISSION-FUEL UPDATE WITH UNDER-RELAXATION. A method that partially overcomes this convergence problem consists in using an under-relaxation. If W_i is the initial weight estimate, and W_{i+1} is the new weight estimate, then the iteration $i+2$ is started with

$$W_{i+2} = \frac{1}{2}\left(W_i + W_{i+1}\right). \tag{15.18}$$

This method does converge but it is considerably slower. A comparison between the two methods is shown in Figure 15.11. Neither option is satisfactory.

C) PREDICTOR-CORRECTOR METHOD. Given an estimate W_f^* of the mission fuel, a mission calculation should predict that the mission fuel is $W_f = W_f^*$. However, if this were to be the case, it would be a lucky outcome. In fact, we have either $W_f > W_f^*$ or $W_f < W_f^*$. The difference $E = W_f - W_f^*$ is the error (or residual). The problem is how to improve this prediction. The algorithm is the following:

- Estimate the mission fuel W_f^* and call the function `MissionFuel(..)`, which returns the fuel W_f. This is the first iteration, $i = 1$.
- If $W_f < W_f^*$, then the initial guess was an under-estimate. Therefore, we perform a new mission calculation with `MissionFuel(..)` and an *increased* fuel

  $$W_f = (1+\epsilon)W_f^*, \tag{15.19}$$

  where ϵ denotes a small fraction of W_f^*. Typical values are $\epsilon = 0.02$ to 0.04, although this depends on how good the first estimate W_f^* is.

- If $W_f > W_f^*$, then the initial guess was an over-estimate. Therefore, we perform a new mission calculation with `MissionFuel(..)` and a *decreased* fuel

$$W_f = (1 - \epsilon) W_f^*. \tag{15.20}$$

- At the end of the second iteration, the error should have decreased because we moved the prediction W_f toward its correct value.
- We keep correcting the mission fuel W_f in either direction (Equation 15.19 or Equation 15.20) until we move from a positive residual E_i to a negative residual E_{i+1} (or vice versa). This event can be verified by the condition $E_i\ E_{i+1} < 0$.
- The correct fuel W_f is then found by linear interpolation between E_i and E_{i+1}.
- A final iteration is carried out to calculate the fuel for each flight segment and the weight breakdown.

For the sake of discussion, assume for example that the initial guess W_f^* is an under-estimate of the solution. Use the underscore "p" to denote the "predicted" fuel, W_{fp}. The algorithm is the following:

$$W_{fp_i} = W_f^* + \epsilon\ W_f^* i, \quad \to W_{f_i}, \quad E_i = W_{f_i} - W_{fp_i} > 0 \tag{15.21}$$

$$W_{fp_{i+1}} = W_f^* + \epsilon\ W_f^*(i+1), \quad \to W_{f_{i+1}}, \quad E_{i+1} = W_{f_{i+1}} - W_{fp_{i+1}} < 0 \tag{15.22}$$

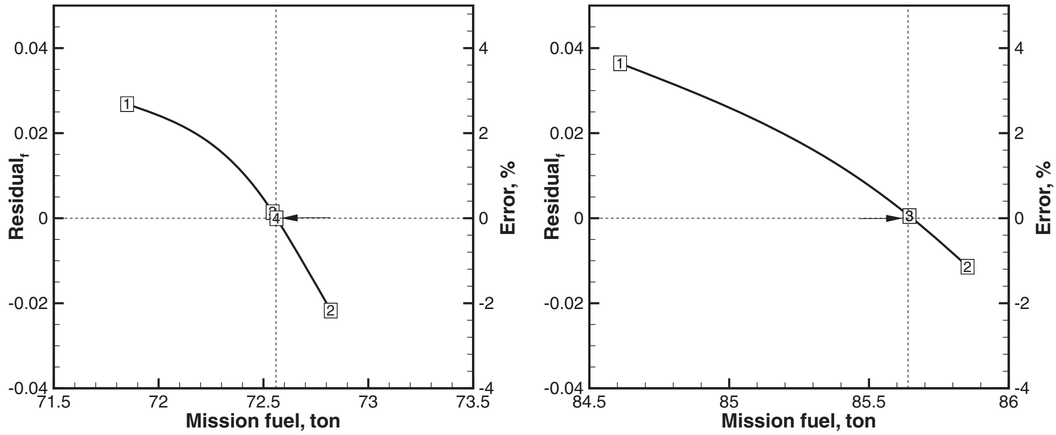
$$W_{fp_{i+2}} = \mathtt{Interpolation}(E_i, E_{i+1}, W_{fp_i}, W_{fp_{i+1}}). \tag{15.23}$$

Equations 15.21 and 15.22 give a prediction of the mission fuel at iteration $i, i+1$, respectively. Equation 15.23 is the correction and the solution of the problem; the corresponding residual is $E_{i+2} \simeq 0$. Note that the residual is calculated between the fuel W_{f_i} returned by the procedure `MissionFuel(..)` and the predicted fuel W_{fp_i}. The minimum number of iterations required for convergence is three.

Figure 15.12 shows the convergence of this method for two values of the required range. This case refers to a Boeing B747-400 with CF6 engines on a standard day, with full passenger load. In graph 15.12a convergence is achieved after four iterations. At the second iteration the residual moves below zero; therefore, the interpolation Equation 15.23 returns a corrected mission fuel. Due to the non-linearity around the solution, another iteration is required. In graph 15.12b convergence is achieved in three iterations.

15.5 Reserve Fuel

A fuel reserve is required by all of the regulators (FAA, ICAO, AEA, and so on). This reserve depends also on the choice of alternate airport and company policies. For example, the Association of European Airlines (AEA) specifies a 200 n-miles (370 km) diversion flight for short- and medium-range aircraft and 250 n-miles (463 km) for long-range aircraft. In addition, a 30-minute holding at 1,500 feet (457 m) altitude and a 5% mission fuel reserve for contingency are required. For domestic flights in the United States it is specified 130 n-miles (~240 km) diversion and 30 minutes holding at 1,500 feet. However, a number of alternatives are allowed, some of which are subject to prior approval. These include a 20-minute extension of the



(a) X = 6,000 km (3,240 n-miles) (b) X = 7,000 km (3,780 n-miles)

Figure 15.12. Predictor-corrector analysis of mission fuel for a Boeing B747-400-CF6; standard day, no wind.

trip or a 3% trip fuel if an alternate airport can be found en route. The ICAO International* rules prescribe that the reserve fuel should take the aircraft to the alternate airport specified in the flight plan and then hold for 30 minutes at 1,500 feet before landing at standard atmospheric conditions. Upon landing, the remaining fuel must be a minimum 3% of the trip fuel.

From an operational point of view, the range may be limited by the absence of diversion airports over a predefined route. The ICAO has a particular rule (ETOPS) permitting twin-engine aircraft to fly longer routes (previously off-limits) that have no diversion airports within a 60- or 120-minute flight. This rule allows several modern twin-engine aircraft to fly over the oceans and in remote parts of the world. A discussion of ETOPS performance with OEI is available in Martinez-Val and Perez[6].

HOLDING STACKS. When holding is required at the destination airport, the aircraft is placed on a *racetrack pattern* that is made of two straight legs and two U-turns (§ 13.2.4). The key aspect of the holding pattern is maximum endurance rather than maximum range. The maximum endurance is achieved at minimum fuel flow or maximum glide ratio. However, at some airports the aircraft may be subject to holding at a specified air speed (or Mach number) that is sub-optimal. For the model Airbus A300-600, the holding speed is 210 kt with a partial slat extension and 240 kt in clean configuration. An alternative is to place the aircraft on a linear holding for part of the time (if the pilots are advised in time). This solution extends the cruise time and reduces the racetrack holding; potentially, it can save a considerable amount of fuel. For example, a 15-minute linear holding of the reference aircraft at $h = 35{,}000$ feet and $M = 0.80$ can save ∼100 kg of fuel.

Optimisation of the holding-pattern performance is critical for short-segment flights because it is not unusual to have a holding time of the same order as the cruise

* ICAO Annex 6: §4.3.6.3.

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time. In the following discussion we fix both the holding Mach number and holding altitude. For additional theoretical aspects of holding optimisation, see Sachs[7].

NUMERICAL METHOD. We carry out the analysis in two steps. First, we find the contingency range. This includes the specified range and the additional range as prescribed by the aviation policies. Then we calculate the fuel corresponding to this range.

- The *diversion distance* X_{div} flown with the maximum landing weight MLW yields an increase in mission range estimated by the equation

$$\Delta X_{div} = (cR)_{div} \frac{W_{mlw}}{W_{to}}, \tag{15.24}$$

  where the coefficient c_{div} accounts for all of the factors in the diversion flight that are sub-optimal (lower speed, altitude, engine efficiency); R (or R_{div}) is the diversion distance.
- The *holding time* (or loiter) flown at the maximum landing weight MLW yields an increase in mission range estimated by the equation

$$\Delta X_{hold} = (cUt)_{hold} \frac{W_{mlw}}{W_{to}}, \tag{15.25}$$

  where the coefficient c_{hold} accounts for loss of efficiency as in the diversion flight. The holding speed is generally half of the cruising speed.
- The *extended duration* of the flight at the cruise speed for a time t_{exd} yields an increase in mission range estimated by the equation

$$\Delta X = Ut_{exd}. \tag{15.26}$$

- The *contingency fuel* is a percentage of the mission fuel $m_{f_{res}}/m_f = 0.05$–0.10, depending on actual policies.

The *equivalent all-out range* is then the sum of all contributions

$$X_{out} = X_{req}\left(1 + \frac{m_{f_{res}}}{m_f}\right) + \left[\, (cR)_{div} + (cUt)_{hold} \right] \frac{W_{mlw}}{W_{to}} + Ut_{exd}, \tag{15.27}$$

where X_{req} is the mission range required. The exact calculation of this parameter is essential. Torenbeek[8] proposed some practical values for the coefficients in Equation 15.27:

$$c_{div} = c_{hold} \simeq 1.1 + 0.5\,\eta_M, \tag{15.28}$$

where η_M is the log-derivative of the propulsive efficiency with respect to the Mach number. For a modern high by-pass turbofan engine this quantity is estimated at $\eta_M \simeq 0.225$ for $M = 0.8$; it increases with the decreasing Mach number, so that $\eta_M \simeq 0.325$ at $M = 0.4$. Equation 15.27 can be further simplified if the reserve fuel is used for extension of the range at the cruise conditions.

An alternative method consists in adding separately the contributions for hold and diversion. Clearly, a number of parameters are required for the solution of this problem: the hold time, altitude and speed; the diversion distance, altitude and speed. The fuel required from these two contingency segments will be added to the mission fuel.

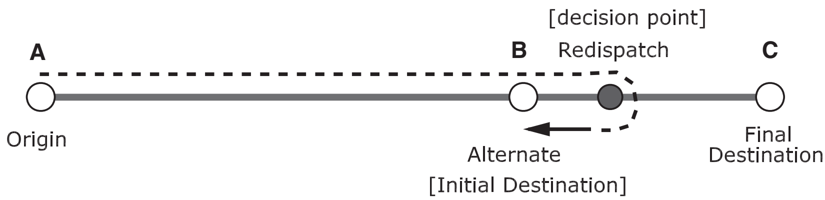

Figure 15.13. Fuel-reserve planning with redispatch.

Finally, we need to consider the *unusable fuel*, that is, the fuel that cannot be pumped into the engines. Although this is a small portion of the fuel capacity (typically 0.4 to 0.8% of the capacity), it is a constant amount of fuel that adds to the operating empty weight.

15.5.1 Redispatch Procedure

The reduction of the reserve fuel relies on the existence of an alternate airport, as shown in Figure 15.13. The graph indicates the departure point (*origin*), the alternate airport (*initial destination*) and the final destination. The graph shows also a waypoint (*redispatch*). The redispatch procedure is regulated by the relevant authorities*.

The reserve fuel must be sufficient to take the airplane from the origin to the initial destination B. In normal circumstances, having passed the point B, the airplane continues its flight; upon reaching the waypoint a decision must be made whether to divert to the initial destination or to continue to the final destination; this is done on the basis of a comparison between the predicted fuel burn and the available fuel. In this scenario, the reserve fuel required is limited to the flight segment AB. By the time the airplane has reached the initial destination, none of the reserve fuel is used, unless the atmospheric conditions turn out to be worse than forecast.

Now consider a more complex network of alternatives, as shown in Figure 15.14, which indicates a refuelling station, an alternative airport beyond the final destination and a second alternate after refuelling.

It is possible to show that the minimum mission fuel with the redispatch procedure is

$$m_f = m_{f_{taxi,A}} + \underbrace{m_{f_{AC}}}_{trip} + \underbrace{0.1 m_{f_{BC}}}_{reserve} + \underbrace{m_{f_{CD}}}_{altern.} + m_{f_{hold,D}} + \cdots \tag{15.29}$$

With the standard procedure, the mission fuel is

$$m_f = m_{f_{taxi,A}} + \underbrace{m_{f_{AC}}}_{trip} + \underbrace{0.1 m_{f_{AC}}}_{reserve} + \underbrace{m_{f_{CD}}}_{altern.} + m_{f_{hold,D}} + \cdots \tag{15.30}$$

The difference between the two procedures is in the reserve fuel. If the redispatch point B is chosen appropriately, a considerable amount of reserve fuel can be saved. Redispatch is not considered in the parametric fuel-reserve analysis carried out in § 15.10.

* See, for example, FAR § 121.631: Original dispatch or flight release.

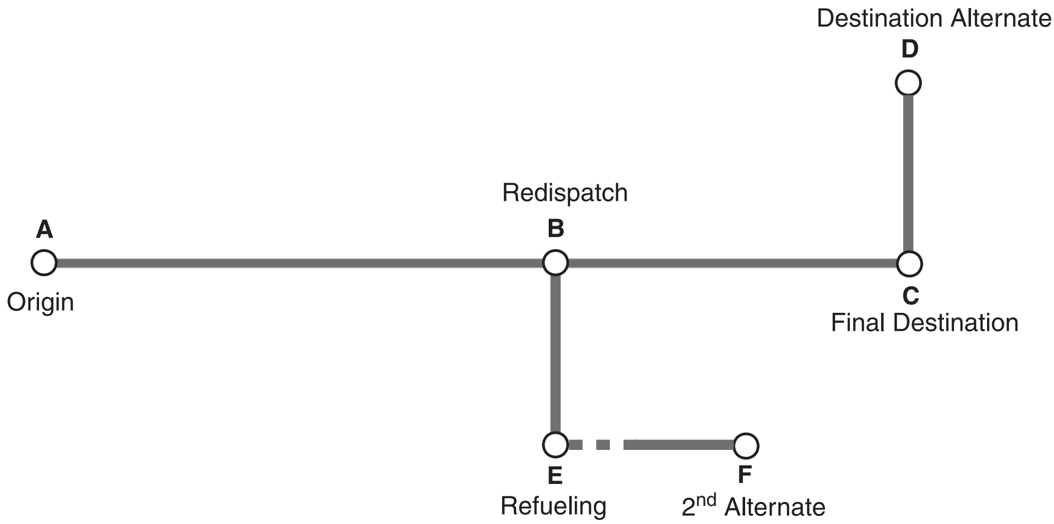

Figure 15.14. Fuel-reserve planning with redispatch for a network of alternatives.

15.6 Take-Off Weight Limited by MLW

Flight over short distances requires a limitation of the TOW due to the relatively low maximum landing weight (MLW). The problem is the following: calculate the distance that the airplane must fly if it is loaded with the maximum payload, takes off with the maximum take-off weight (MTOW) and lands with the MLW. This range is calculated from the following numerical procedure:

- The aircraft starts with MRW, a payload W_{msp} = MSP, and whatever fuel W_f can be allocated, so that

$$\mathrm{GRW} = W_e + W_{msp} + W_f. \tag{15.31}$$

- The fuel required to taxi out, take off and climb to a cruise altitude (to be determined) is calculated and added. By the time the aircraft has reached the "cruise" altitude, it will have burned a fuel W_1 and travelled a distance x_1.
- Now consider the descent phase. Estimate the weight at the top of descent from

$$W_d \simeq W_{mlw}(1 + r), \tag{15.32}$$

where r denotes the fuel reserve, as percent of the initial fuel. Calculate the descent fuel W_3 and the en-route descent distance x_3.
- Finally, consider the cruise range x_2 (unknown). The aircraft weight at the start and end of the cruise should be

$$W_{start} = \mathrm{GRW} - W_1, \qquad W_{end} = W_{start} - W_{f_c}, \tag{15.33}$$

where the cruise fuel W_{f_c} is

$$W_{f_c} = W_{mtow} - W_{mlw} - W_1. \tag{15.34}$$

- Estimate the cruise range from the range equation

$$x_2 = \frac{U}{g f_j}\left(\frac{L}{D}\right)\log\left(\frac{W_{end}}{W_{start}}\right). \tag{15.35}$$

- The MLW-limited range is the following:

$$x_{mlw} \simeq x_1 + x_2 + x_3. \tag{15.36}$$

The range estimate given by Equation 15.36 can be improved by further iterations, but the result does not change a great deal.

As a practical example, consider the Airbus A-380-861 with GP-7270 engines. This aircraft has the following characteristics: MSP = 90.7 tons; OEW = 270.3 tons, MLW = 386.0 tons. If we load the airplane with the MSP, we have ZFW = 361.0 tons, which is exactly the certified MZFW. In this case, the amount of fuel required to reach the MLW is $\sim$27 tons. With such an amount of fuel, the airplane can quickly return to the airfield of origin if it was to abort the mission. Because in a typical climb the aircraft burns $\sim$9 tons of fuel, in addition to $\sim$ 2 tons for taxi and take-off, the aircraft will have to dispose of about 16 tons of fuel before landing.

15.7 Mission Problems

There are quite a variety of mission scenarios. Even in the relatively simple case of a transport aircraft, important decisions may have to be taken when planning a flight and during the flight. In this section we deal with a number of practical problems. We begin with the mission analysis with an intermediate stop (§ 15.7.1); in this case we want to understand whether there are benefits in terms of fuel consumption if we split the mission and call at an intermediate destination. The second problem deals with fuel tankering, which is a practice sometimes used to save money at destination airports if the fuel is more expensive there (§15.7.2). Finally, we consider a contingency scenario in which the pilot may have to abort the flight and decide whether to return to the origin airport or continue to its planned destination (§15.7.3).

15.7.1 Cruise with Intermediate Stop

Is it more economical to fly a long distance no-stop or with an intermediate stop? Some airlines occasionally offer cheaper seats to a destination via an off-route stop, which is counter-intuitive to both the customers and the airplane designers. The aviation industry has developed aircraft for very long range to serve distant points without en-route stops. Some of the current-generation airlines can fly, in some limited scenarios, in excess of 10,000 n-miles. This is close to the ultimate *global range*, 20,000 km ($\sim$10,790 n-miles), advocated by Küchemann[9]. The global range is half the Earth's circumference at the equator and would allow an aircraft to fly from any point to anywhere around the world – at least in principle.

The problem is one of long-range cruise in which a subsonic commercial jet has to travel from airport A to a destination airport B. The distance along the flight corridor is within the certified maximum range of the aircraft. However, here we consider the problem of an *en-route* stop at airport C. The aircraft carries enough fuel to reach airport C, refuels at this base, takes off again and flies to its final destination B. We do not consider the costs of landing at C or the direct operating costs incurred by the operator for increasing the time of the return journey. If x is the distance to be flown in a direct flight, and x_1, x_2 are the flight segments, we assume that

$$x_1 + x_2 = x. \tag{15.37}$$

Table 15.1. *Fuel use for mixed long- and short-range service of the Boeing B777-300 (calculated)*

X[n-m]	m_f[kg] total	m_f[kg] climb	m_f[kg] cruise	$\dot{m}_f$[kg/s] cruise
300	6,674	2,553.9	955.0	2.04
5,000	106,287	3,056.3	94,864.6	2.63
5,300	112,396	3,056.6	100,662.7	2.62

In other words, the en-route stop C is on the route to the final destination B. A stop with a diversion clearly increases the total range. The method consists in calculating the mission fuel for three flight segments: x, x_1, x_2. We then calculate the relative fuel cost. This is the ratio between the fuel required by the flight with en-route stop and the direct flight:

$$m_{fr} = \frac{m_f(x_1) + m_f(x_2)}{m_f(x)}. \tag{15.38}$$

This quantity can be plotted as a function of x_1/x (the ratio between the first flight segment and the total range). Consider the case in which an airline operates an intercontinental service between A and B separated by 5,000 n-miles. Another important destination is C, which is 300 n-miles from B. The results of the calculation performed with the `FLIGHT` program are shown in Table 15.1. The data show that stopping at en-route airport at 5,000 n-miles and proceeding to the final destination 300 n-miles away causes an increase in fuel consumption by 0.5%.

15.7.2 Fuel Tankering

The aircraft has to carry the minimum required amount of fuel to perform its mission. However, this type of analysis does not take into account any contingencies arising from the costs of purchasing fuel. Some commercial operators use the practice of "tankering", that is, loading more fuel than required by a mission profile to offset the costs of purchasing fuel at a higher price at destination. As a rule, the longer the flight, the less capacity there is for tankering. Figure 15.15 shows the tankering performance of the Airbus A320-200, adapted from the manufacturer's data*. The optimum weight is not fully specified; that is, it is not clear what payload is carried.

Assume that the required range is X_{req} for a total payload weight W_p. The fuel required for this mission is W_f (including fuel reserves). To establish the optimum tankering, we need to forecast the payload and the weather for the return journey. At this point the matter can become complicated; to simplify, we assume that the aircraft performs the return journey with the same payload and atmospheric conditions. Hence, the amount of fuel required is the same.

From the point of view of the calculation, there is no difference between fuel weight and payload weight. The increase in fuel burn due to an increase in payload (or fuel itself) is the derivative

$$m = \frac{dW_f}{dW_{to}}. \tag{15.39}$$

* Airbus: *Getting to Grips with Fuel Economy*, Issue 3, Blagnac, France, July 2004.

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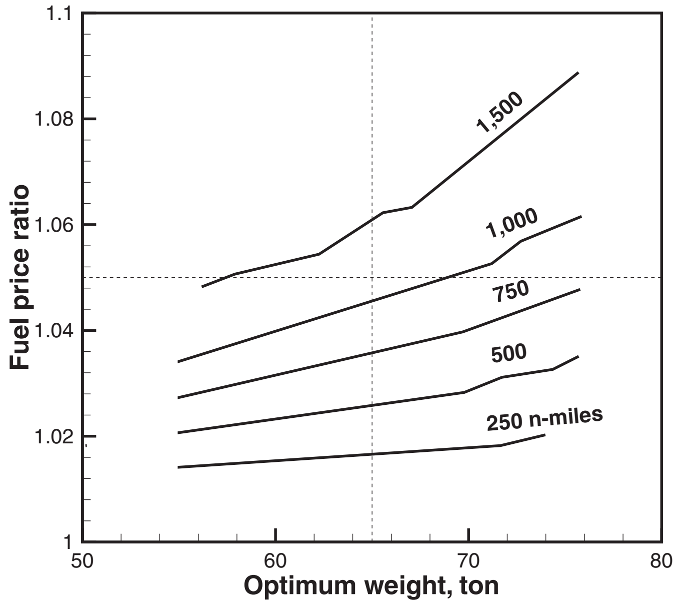

Figure 15.15. Tankering performance of the Airbus A320 relative to cruise FL-330. Data adapted from Airbus.

This derivative is always positive. Therefore, the *additional* fuel required to carry the tanker fuel $W_{f_{tanker}}$ is $mW_{f_{tanker}}$; the total increase in take-off weight (or increase in fuel weight) is

$$\Delta W_{to} = \Delta W_f = (m+1) W_{f_{tanker}}. \tag{15.40}$$

The price to be paid at departure for this additional fuel is

$$\mathcal{P}_d = m p_d W_{f_{tanker}} + p_d W_{f_{tanker}}. \tag{15.41}$$

The first contribution is due to the cost of carrying the tanker fuel; the second term is the cost of the tanker fuel itself. The cost saved at arrival is

$$\mathcal{P}_a = p_a W_{f_{tanker}}. \tag{15.42}$$

The difference $\mathcal{P}_d - \mathcal{P}_a$ must be negative (which corresponds to a saving):

$$\mathcal{P}_d - \mathcal{P}_a = m p_d W_{f_{tanker}} + p_d W_{f_{tanker}} - p_a W_{f_{tanker}} < 0, \tag{15.43}$$

which implies

$$\frac{p_a}{p_d} > 1 + m. \tag{15.44}$$

The tanker fuel in Equation 15.43 is indeterminate. Note, however, that the fuel derivative (Equation 15.39) increases with the tanker fuel. Therefore, for a given favourable fuel price ratio, it becomes progressively less convenient to tanker any fuel; alternatively, for a marginally favourable fuel price ratio, an increase in tanker fuel can be offset by the additional cost of carrying it. Mathematically, the optimum tanker fuel is $W^*_{f_{tanker}}$ such that

$$m = \frac{p_a}{p_d} - 1. \tag{15.45}$$

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For our calculation, we fix the price ratio $p_a/p_d > 1$ (or $p_d/p_a < 1$). The computational procedure is the following:

1. Specify the required range, payload and fuel price ratio.
2. Establish a minimum and a maximum tanker fuel $W_{f_{tanker-1}}$, $W_{f_{tanker-2}}$.
3. Use the bisection-iteration method between these two values of $W_{f_{tanker}}$ to calculate the fuel derivative m that satisfies Equation 15.45. This procedure should converge to $W^*_{f_{tanker}}$ (with some caveats).

With reference to point 3 in the previous procedure, there is an inner loop that calculates the fuel sensitivity by using central differences, that is, two mission calculations with the tanker fuel $W_{f_{tanker}} \pm dW_f$, where dW_f is a small number.

For certain values of the fuel price ratios, the procedure does not converge. In fact, if p_a/p_d is high, the procedure attempts to calculate the tanker fuel corresponding to a high value of the fuel derivative (Equation 15.45), which is incompatible with the flight systems. For example, if the fuel price at arrival is double the cost at departure, the fuel derivative that solves Equation 15.45 should be $m = 1$, a far higher value than actually required. A similar difficulty exists for $p_a/p_d \simeq 1$, which implies that for marginal differences in fuel costs, it is uneconomical to tanker any fuel. Realistic values of the fuel derivative are $m = 0.01$ to 0.07, depending on aircraft, range, payload and other factors. We conclude that in these circumstances it is always convenient to use tankering.

For the Airbus A320-200, we have carried out the following analysis: $X_{req} = 750$ n-miles; bulk payload $W_p = 2{,}000$ kg; passenger load equal to 80%; standard day, with a price fuel ratio $p_a/p_d = 1.05$. The optimum tanker fuel is $W_{f_{tanker}} = 1{,}273$ kg, corresponding to a ramp weight of ~65,400 kg. This result does not match the value quoted by the manufacturer, which in any case is not fully specified. The iterative analysis carried out with the `FLIGHT` program is shown in Figure 15.16. The error on Equation 15.45, that is, the difference between left- and right-hand sides, is ~0.002 after four iterations.

Because fuel tankering causes additional fuel burn, it also causes additional environmental emissions. If environmental costs are added to the analysis, the incentive to tanker may disappear altogether. In any case, to be precise we would need to account for secondary costs, such as increased maintainance costs on engines, tyre and brakes wear, discounted for the shorter turn-around time at destination. Thus from an operational point of view, it may be more convenient to formulate the problem as a break-even fuel price ratio.

15.7.3 Equal-Time Point and Point-of-No-Return

If diversion is required due to external events, landing must be planned at an airport different from the planned destination. Such events may include engine failure, loss in cabin pressure, adverse weather, air traffic control and more. A contingency plan is formulated on the basis of the equal-time point and the point-of-no-return. The situation is shown in the sketch of Figure 15.17. These quantities are defined as follows:

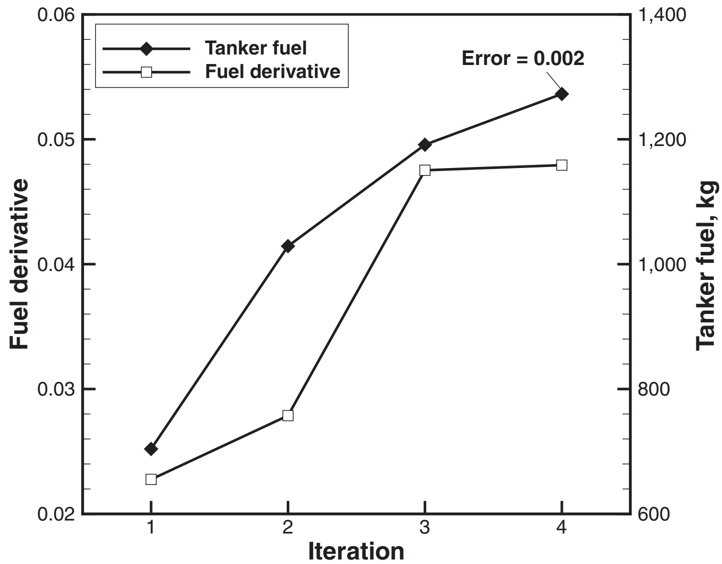

Figure 15.16. Fuel-tankering calculation for the Airbus A320-200.

- The equal-time point P_{ET} is the point during the flight when the time required to return to the airport of origin and the time required to continue the flight to the planned destination are equal.
- The point-of-no-return P_{NT} is the point during the flight when the fuel remaining is not sufficient to return to the airport of origin.

The point-of-no-return comes *after* the point of equal time. The wind direction is critical. These two points must be converted to a geographical position and must be assigned an estimated time of arrival.

Assume that the aircraft has reached the point P_{ET}. Due to external events, it must be decided whether to return or continue. At this point, the fuel required for the inbound flight (return to origin) is less than the fuel required for the outbound flight. However, the flight time is the same. In this case, there is a time constraint (or priority) rather than a fuel constraint.

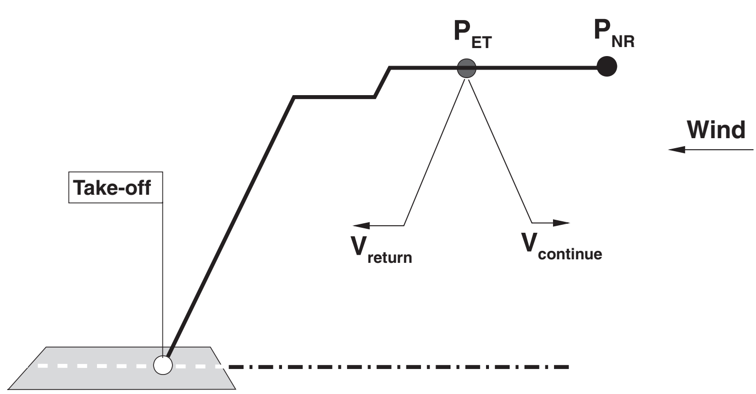

Figure 15.17. Point of equal time and point-of-no-return.

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Next, assume that the aircraft has reached the point P_{NT}. The fuel required to return would be more than the fuel required to continue. In this case, there is a fuel constraint rather than a time constraint.

If we use our `FLIGHT` program, the calculation of P_{ET} and P_{NT} is relatively straightforward if there is no wind. We calculate a full mission with the procedure `MissionFuel(..)`, which returns a mission time t and a mission fuel m_f; we then calculate *a posteriori* P_{ET} as the point where the flight time is $t/2$; P_{NT} is the point where the fuel burned is $m_f/2$. However, this is not a satisfactory solution for two reasons. First, the wind cannot be neglected. Second, we cannot wait until the end of the flight to decide what to do in an emergency situation. When time is critical and the aircraft must return to the origin, we must consider the time required to make a U-turn (§13.2.4). A procedure to take into account these effects is the following:

1. Specify the mission parameters, including range, wind and wind direction.
2. Perform the mission analysis with function `MissionFuel(..)`.
3. Estimate P_{ET} as a point somewhere in the cruise segment, determined by a distance X_{ET} from the origin airport and a flight time t_{ET}.
4. Perform the mission analysis with the function `MissionFuel(..)` up to X_{ET}; the distance to continue would be $X_c = X_1 - X_{ET}$; the remaining time would be $t = t_1 - t_{ET}$.
5. Perform a U-turn at the cruise altitude (§ 13.2.4).
6. Switch wind direction.
7. Continue the mission to the destination (e.g., for the remaining distance).
8. Compare the time to return t_{ET} with the time to continue t: if $t_{ET} > t_c$, then the point P_{ET} was over-estimated; otherwise, it was under-estimated.

The last point of this procedure can be implemented into a bisection method that iterates between an under-estimate and an over-estimate of the solution. The procedure can be computationally demanding because it requires two calls to `MissionFuel(..)` to bracket the solution and five to seven bisection iterations. Because each mission analysis requires about four to five iterations to converge, there are up 40 to 50 calls to `MissionFuel(..)`, unless the converge criterion is relaxed. The aircraft will have some mandatory fuel reserve (§ 15.5). In any case, we assume that the aircraft maintains the fuel reserve.

An example is the following. A model Boeing B747-400 with CF6 engines is on a mission from London Heathrow to New York Kennedy airport. The nominal required range is 5,700 km (3,078 n-miles), with 1 metric ton of cargo and 80% passenger load. It encounters headwinds of 39 kt at cruise altitude. For this specific case, the equal-time point is estimated at 1,615 n-miles (52.4% of the required distance) at a flight time of 233.5 minutes from take-off.

15.8 Direct Operating Costs

Direct operating costs (DOC) are the costs incurred in the ownership and operation of an aircraft. They include the cost to operate scheduled and unscheduled flights, to insure and to maintain the aircraft as airworthy. The aircraft costs money even if it stays on the ground. The parametrisation of the fuel costs is aleatory, due to the costs of aviation fuel on the international markets and to the cost of fuel at different airports around the world.

An example of DOC analysis is given by Beltramo *et al.*[10], who developed cost and weight estimating relationships for commercial and military transport aircraft. Kershner[11] has demonstrated that whilst DOC have consistently decreased over time, the impact of the fuel cost has remained high and in some historical circumstances has increased to more than 50%. The prediction of the operational costs due to fuel consumption can be calculated with the methods explained in this chapter. Other studies in this area focus on the fuel consumption for subsonic flight[12] and for supersonic transport[13].

The appearance of *budget airlines* on the major world markets has contributed to a substantial change in the structure of the DOC. It is sometimes exciting to learn that we can buy a ticket to an international destination within Europe or the continental United States at a cost comparable to the cost of this book. Cost items such as ticketing, customer service, seat allocation, baggage handling, ground services, in-flight catering, airport taxes, leasing contracts and so on have been dissected, reduced or removed altogether from the DOC. However, the cost of the fuel remains essentially the same. This cost is an essential part of the aircraft performance.

CALCULATION METHOD. The calculation of the DOCs of an airplane is a contentious issue. Data from major airlines are published by the IATA and are closely monitored. On the design side, there are unsubstantiated claims made by the manufacturers. On the operational side there are uncertainties arising from market conditions, financial arrangements and a long list of externalities (inflation, variable interest rates, fuel costs, tax liabilities, labour relations, adverse weather, and so on). Thus, if we want to be realistic, we need to build these uncertainties in the costs analysis. With the foregoing cautions in place, we provide a simplified analysis that is based on about half a dozen major cost items. These are:

1. *Ownership costs*, which are a combination of cash payments on acquisition, loan repayment over a fixed length of time and residual value of the airplane at the end of the financing contract.
2. *Insurance costs*, which will be considered a fraction of the value of the airplane, although this might not be the case (the third-party liabilities do not decrease over time).
3. *Fuel costs*, based on average costs between departure and arrival at *today's price, adjusted for inflation*, or an agreed deviation from it.
4. *Crew costs*, consisting of pilots and flight attendants, based on full-time rates.
5. *Spare parts costs*, consisting in parts for the propulsion system, airframe, landing gear and any other items.
6. *Labour costs*, based on recognised rates (in today's prices, adjusted for inflation) for propulsion, in-house and contracted-out operations; part of the problem is to gather reliable data for these items, the man-hours required to carry out the service and the down-time for the airplane.

7. *Other costs*, including landing fees, ground handling, pilot training, on-board service items, costs for off-station residence and so on. None of these costs can be neglected because they make up the difference between being profitable and making a loss.

We start with the ownership costs, which are the most onerous. These costs are estimated from the equation

$$C_1 = \frac{1}{n}\left[\mathcal{P} - F(1+I)^n - R_n\right], \tag{15.46}$$

where $\mathcal{P}$ is the aircraft price agreed at the acquisition date; F is the financing (i.e., the fraction of the acquisition price requiring a loan); I is the interest rate on the loan, assumed to be constant over the life-time n (in years) of the loan itself; and R_n is the residual value after n years. The latter item is uncertain and must be based either on today's price or perhaps corrected for inflation at the end of the loan. To be on the safe side, we could assume that the aircraft will be written off at the end of the period, $R_n = 0$. The insurance cost is estimated from

$$C_2 = R_j \frac{I}{100}, \tag{15.47}$$

where R_j is the residual value at year j. The fuel cost over one year is

$$C_3 = N\overline{C}_{fuel} m_f, \tag{15.48}$$

where N is the number of cycles, $\overline{C}_{fuel}$ is the average fuel cost per kg, and m_f is the fuel burn per cycle over the specified stage length. This quantity is calculated with a mission analysis and will depend on a realistic estimate of the passenger-load factor and gross take-off weights. Fuel-price oscillations are accounted for in the averaging. Increases in fuel consumption due to engine age are accounted for with a separate model (see, for example, § 5.3.6). The deterioration-adjusted fuel burn is

$$D_f(x) = \alpha_1\left[1 - e^{-\alpha_2 x}\right] + (n_w - 1)/c, \tag{15.49}$$

where $x = N/1{,}000$ is the number of thousand cycles since the last wash, α_2 a deterioration lapse ($\alpha_2 \simeq 1$ to 1.5) and α_1 is the deterioration effect on the fuel burn after 1,000 cycles ($\alpha_1 \simeq 2$ to 3); n_w is the number of total engine washes and c is the irretrievable loss of performance over time ($c \simeq 20$). The crew cost is

$$C_4 = n_1 S_1 \left(\frac{T_b}{T_o}\right)_1 + n_2 S_2 \left(\frac{T_b}{T_o}\right)_2 + n_1 S_3 \left(\frac{T_b}{T_o}\right)_1, \tag{15.50}$$

with n_1 and n_2 the number of pilots and average flight attendants required to operate the aircraft year round; S_1 and S_2 are the corresponding salary costs based on full-time service, T_b is the total number of utilisation hours per year, and T_o is the contractual number of working hours. The number of flight attendants will be estimated from the passenger load and the type of services (about 1/30 in economy and 1/10 in premium/business class). Other major crew-related costs include pilot training S_3, which takes place ahead of the acquisition and at scheduled intervals during the lifetime of the airplane. The costs of the spare parts is

$$C_5 = \sum_k \mathrm{SP}_k, \tag{15.51}$$

where SP_k $(k = 1, \cdots)$ denotes spares for the propulsion system, APU, airframe, landing gear, tyres and any other items. The maintenance schedule depends on the number of cycles and airplane age; it is divided into three broad categories: propulsion and associated systems, in-house and contracted out. Maintenance costs increase with ageing aircraft and cannot be oversimplified. As a result, the labour costs associated to maintenance are likely to go up as the number of hours required increases. The associated maintenance costs are:

$$C_6 = \sum_k \mathcal{L}_k \mathrm{MH}_k, \tag{15.52}$$

with $\mathcal{L}_k$ the labour rates (any currency per hour) for the categories listed herein and MH_k the number of man-hours required in each category to carry out the maintenance schedule. The number of man-hours involved depends on the airplane (gross weight, number of engines, landing gear) and the number of cycles. These costs suffer a jump as soon as the airplane comes off the warranty period. In practice, this means that part of the costs are transferred from the manufacturer to the operator. This transfer is called *burden* and is not necessarily an ageing cost.

The landing fees depend on the airport, the weight class of the aircraft and local congestion; it may be subject to additional levies, such as CO_2, NO_x emissions and noise emissions. We assume that there are two major contributions: from the gross weight (or number of seats) and the overall emissions, with weights c_1 and c_2, respectively:

$$C_7 = 2(c_1 W + c_2 E)N. \tag{15.53}$$

As anticipated, the maintenance costs increase over time. Figure 15.18 shows such effects. These charts have been elaborated from a statistical analysis carried out by the Rand Corporation[14]. The data have been generated from a limited sample of airplanes analysed; their limitations are discussed in the report cited. The charts provide some weights to account for the costs of major items, such as airframe, engine and burden effects. In all cases, it is shown that new aircraft have rapidly increasing costs, which eventually flatten out with age. Apart from having to update the coefficients, depending on type of aircraft, technology level and other factors, the conclusions are generally valid. If we use the form

$$C_{56} = C_{5,1} + C_{6,1} \tag{15.54}$$

to denote the combined parts and man-hour costs for the engine, the function C_{56} should follow the trend of Figure 15.18b. Even then, we need to establish values for labour rates $\mathcal{L}$ and man-hours MH at year zero. The DOC model, as presented, requires the evaluation of about 30 independent parameters. A summary of these parameters is in Table 15.2 for clarity.

On a final note, we report that there has been an introduction of leasing contracts for the propulsion system. These contracts allow the airline operator to simply pay a cost proportional to the number of flight hours. When this is the case, the analysis proposed here will have to be adjusted to take into account this different cost structure.

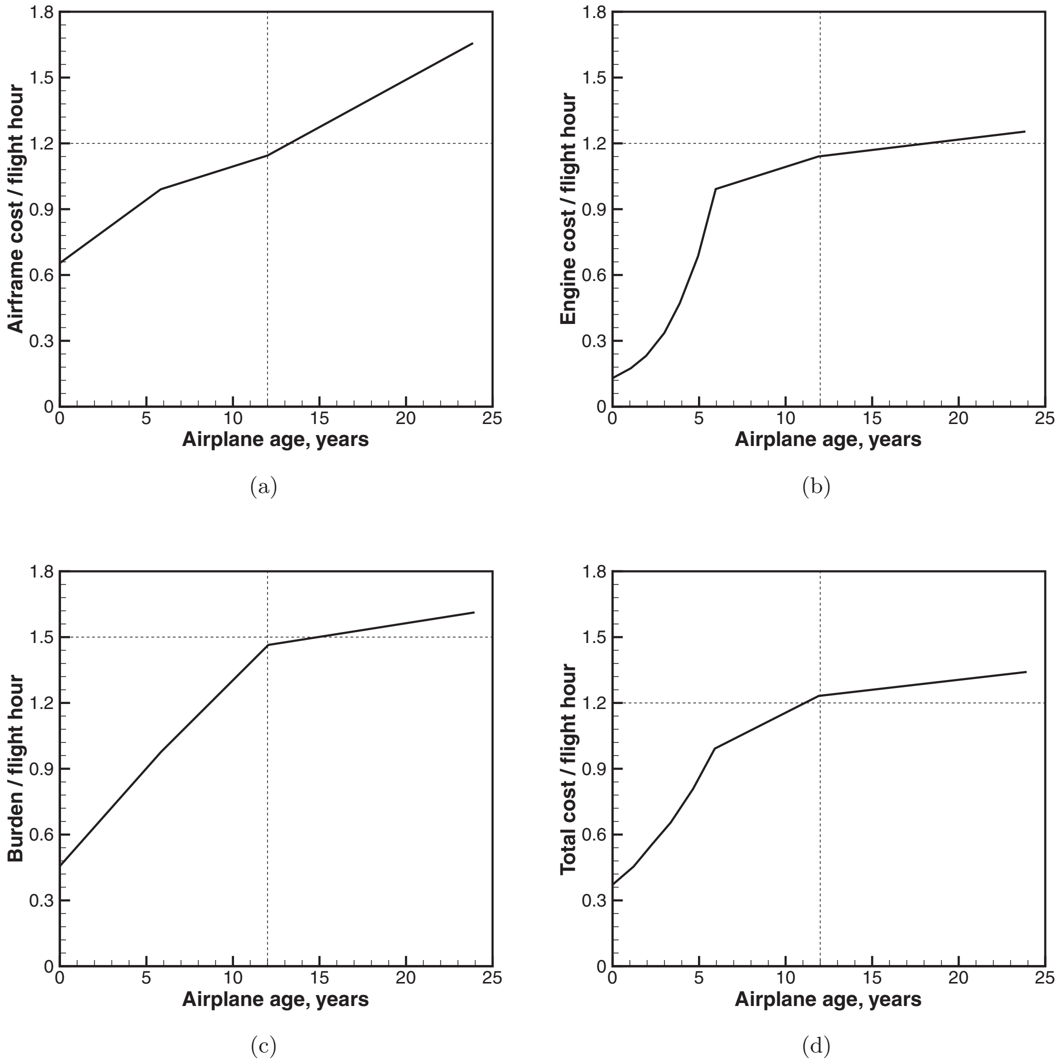

Figure 15.18. Estimated ageing effects on maintenance costs.

Table 15.2. *Summary of parameters for DOC model*

Item	Parameters	Reference
Ownership	$\mathcal{P}$, F, i, n, R_n	Equation 15.46
Insurance	I	Equation 15.47
Fuel	N, $\overline{C}_{fuel}$, m_f	Equation 15.48
Crew	n_1, n_2, S_1, S_2, T_o	Equation 15.50
Spare parts	SP_1, $\mathrm{SP}_2 \cdots$	Equation 15.51
Labour rates	$\mathcal{L}_1$ $\mathcal{L}_2 \cdots$ MH_1, $\mathrm{MH}_2 \cdots$	Equation 15.52
Landing fees	c_1, c_2, E	Equation 15.53
Other costs		To be specified
Ageing effects on fuel	α_1, α_2, n_w	Equation 15.49
Ageing effects on maintenance		Figure 15.18

Table 15.3. *Calculated payload fuel efficiency for long-haul commercial flight; all flights at* $M = 0.80$*; all weights are in metric tons*

Aircraft	OEW	MTOW	PAY	X [n-m]	Fuel	GTOW
B	129.9	275.0	35.0	7,560	74.62	244.1
A	90.1	165.0	35.0	3,780	55.69	156.1
B	129.9	275.0	50.0	7,560	77.94	263.1

PRODUCTIVITY MEASURES. There are specific parameters used by commercial operators to define their productivity. One of the most important is the Available Seat per kilometre (ASK) or nautical mile (ASM). This parameter expresses the number of seats available for sale, flown 1 km (or 1 n-mile). Then there is the Cost per available seat-km (CASK) or seat-n-mile (CASM). The CASK is equal to ASK/DOC and determines the actual cost incurred by the operator of making a seat available on a certain airplane over a certain route. Because the DOC changes as a function of the various parameters, so does the CASK. Sometimes a cost analysis is carried out by excluding the cost of fuel; this operation requires the ex-fuel CASK. Finally, there is the revenue per ASK (RASK). To make a profit, a commercial operator must have RASK > CASK.

15.9 Case Study: Aircraft and Route Selection

How do we select a new airplane from two or more competing airplanes? We will have to face a mountain of marketing documentation about the benefits of one airplane with respect to the other; each airplane is inevitably 20% more fuel efficient than the other one, with the costs of the competing airplane being calculated by the competitor, not the manufacturer. This is not an easy decision to make and even between two competing airplanes of one manufacturer, a business decision will have to be made on the basis of hard evidence; the process can be costly and time-consuming.

In the next analysis we will refer to the relatively simple problem of deciding which airplane to dispatch for a certain mission. We assume that we have to connect two airports separated by a distance of 14,000 km (~7,560 n-miles) and that we have to deliver a payload of 35,000 kg. We have airplanes A and B, whose OEW and MTOW are given in Table 15.3.

From the payload-range chart, we conclude that aircraft B can make a no-stop flight to destination. For this aircraft, we estimate a fuel burn of ~77.62 tons. The service can also be done with aircraft A with an en-route stop half-way (7,000 km; 3,780 n-miles). The total fuel required (including the two legs) in this case is ~55.69 tons, which is a savings of about 25%. Therefore, the second option is more economical. If, instead, we are required to deliver a 50-ton payload, aircraft A is not large enough; aircraft B instead can easily accommodate that payload, and the increase in fuel consumption over the previous case is only ~4.5%. The summary of our calculations is in Table 15.3.

Figure 15.19 shows the result of a parametric analysis carried out with a model Boeing B777-300 airplane with GE-92 engines. The purpose of this analysis was to

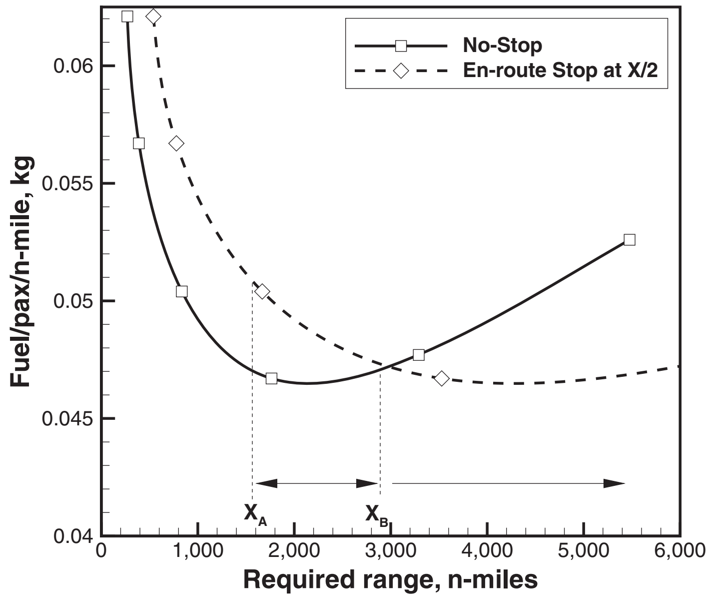

Figure 15.19. Analysis of B777-300 airplane model flight with and without en-route stop; standard day, no wind.

understand how the en-route stop affects the fuel cost. The results indicate that for a distance above ∼3,000 n-miles, an en-route stop could save fuel. The reason for this is that the longer the required range, the heavier the airplane, as a result of having to carry a larger amount of fuel. The solution shown in Figure 15.19 is not as straightforward as it may look. First, the parametric analysis is carried out with the assumption that a suitable airport is available at exactly half the distance. Second, there can be considerable costs in landing at an intermediate destination. Third, for some values of the range, it would not be possible to stop anyway. For example, for a 3,000 n-miles flight it would make no sense to stop 500 n-miles to the final destination unless there is a commercial reason to do so. Refer also to the case illustrated in Table 15.1. The range X between X_A and X_B is the most efficient and corresponds to a fuel burn (per passenger per n-mile) that is within 1% of the minimum. The results shown refer to full passenger load. Therefore, the fuel/pax/n-mile is equivalent to fuel/seat/n-mile. The result would be slightly different for a less-than-full aircraft.

The solution can be further refined to include other factors and possibly an estimate of the DOC. Green[15] showed an analysis for cargo aircraft and recommended that from the point of view of fuel efficiency, a flight segment shall not exceed 7,500 km (∼4,000 n-miles) at the current level of technology. In-flight refuelling is an option that can be considered for either extending the range or increasing the payload for a fixed range. However, the calculation must include the cost of operating a tanker aircraft and to maintain a loiter trajectory for some time[16].

The next analysis focuses on the fuel use as a function of the required range. Figure 15.20 shows the fuel fraction (with respect to the fuel embarked) used for climb to the ICA and for cruise. The cruise fuel converges toward 80% of the total

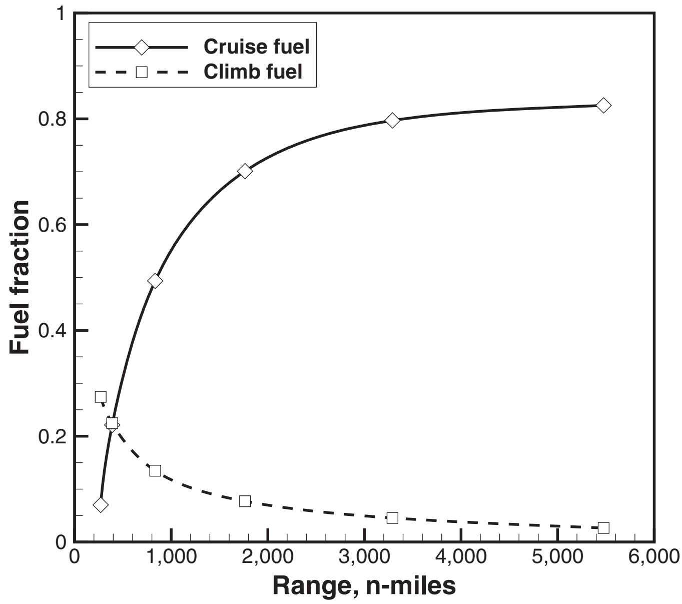

Figure 15.20. Analysis of fuel use by the Boeing B777-300 with GE engines; standard day, no wind; full passenger; no bulk cargo.

fuel embarked, whereas the climb fuel decreases to a few percent. Shorter journeys require relatively higher portions of the fuel to be used for climb. Specifically, on a 250-n-mile journey the climb fuel exceeds 25% of the total fuel embarked. The remaining fuel is used in the other flight segments and includes the reserve fuel.

15.10 Case Study: Fuel Planning for Specified Range, B777-300

We apply the methods presented earlier to the calculation of the mission fuel and gross take-off weight for the Boeing B777-300 with General Electric GE092 turbofan engines and 331-500 APU. A summary of mission calculations is shown here, which are computer outputs from the program `FLIGHT`. Looking at a more detailed level, we have reports for the various segments of flight. Table 15.4 shows a list of flight conditions and assumptions. Table 15.5 is a summary of fuel/weight planning resulting from the calculations. We discuss separately the different flight segments.

Taxi-Out Report

It is not unusual that the aircraft must remain on hold waiting for a departure slot, due to local traffic or bad weather. Fuel consumption on the taxi-way can be so high that the flight must be aborted. The jet engines' performance is optimised for cruise, and considerable efficiency penalties are faced on the taxi-way. Therefore, taxi-times must be estimated with accuracy.

Some economy can be gained by taxiing the aircraft with OEI or by towing the aircraft to and out of the gate, or by parking the aircraft away from the gate. The

Table 15.4. *Operational data for mission analysis in case study*

Item	Value	Unit
Required range	2,698	[n-miles]
Required bulk payload	0.0	[ton]
Required pax	74.6%	full capacity
Outbound taxi time	10	[min]
Inbound taxi time	8	[min]
Wind speed at cruise	−5.0	[m/s]
Change in temperature	0.0	[K]
Take-off airfield altitude	50.0	[m]
Landing airfield altitude	50.0	[m]
Derating	none	
Hold altitude	507	[m]
Hold time	30.0	[min]
Diversion distance	200.0	[n-miles]
Baggage allowance/pax	15	[kg]
Average pax weight	80	[kg]

OEI procedure is not recommended in a number of cases, including high GTOW, slippery ground, nose-wheel steering performance, limitations on jet blasts, safety for ground personnel and other reasons. It can be more efficient to taxi at a high speed (although the limit is set to 25 to 30 kt) with a burst of engine power, and then run the engine with idle power.

A first-order estimate of the taxi fuel is found with the method shown in § 9.10 (Chapter 9). A summary of the taxi-out performance for this specific case is shown in Table 15.6. This table indicates that a considerable amount of fuel, half a ton, is required to reach the start of the runway! Thus, we face an important operational and environmental problem that must be addressed by more efficient procedures.

Take-Off Report

This report is shown in Table 15.7. There is a large set of airplane and operational data required for the complete take-off analysis. The wind speed and runway state are specified on input. The remaining data are calculated by the computer program. The flap setting is also part of the solution, as the numerical algorithm attempts to perform a take-off with the minimum flap setting compatible with a minimum initial climb gradient. Included in the analysis is the result of the thermo-structural model for the tyres (§ 14.5) and the minimum control speed on the ground (§ 9.6).

En-Route Climb Report

Key data are the time to climb, the distance to climb and the fuel to climb. The various climb segments are identified and report data such as average climb rate, fuel use and fuel flow. Landing-gear retraction and flaps retraction are done instantaneously. Climb is done in four segments, the last one being a constant-Mach climb to the

Table 15.5. *Summary of flight-planning analysis*

Fuel Use Report

Segment	m_f [kg]	m_f [%]	X[n-m]	t [min]	Notes
Climb	4,230.2	8.05	191.76	28.4	
Cruise	37,415.5	71.16	2,317.72	299.7	
Descent	5,037.1	9.58	210.44	37.3	
Take-off	214.6	0.41			
Taxi-out	504.1	0.96	1.12	10.0	
APU/ECS	4,674.9	8.89			
Taxi-in	415.9	0.79			from reserve
Fuel burn	52,582.0				

Fuel Breakdown Report (all data [kg])

Taxi-out	504.1	
Take-off	214.6	
Climb	4,230.2	
Cruise	37,415.5	
Descent	5,037.1	
Approach	89.5	
APU/ECS	4,674.9	
Diversion/Hold	2,498.1	
Reserve	2,754.0	
Total	57,834.1	42.1% of full tanks

Final Weight Report (all data [kg])

OEW	161,355
Bulk payload	0.000
Passengers	23,520
Flight crew	1,140
Useful payload	24,520
Service items	0.873
Fuel	57,834
Non-usable fuel	824
Ramp weight	243,658
BRGW	243,658
Zero-fuel weight	187,887
Take-off weight	249,816
Final Cruise weight	239,096
Landing weight	193,075

optimal (or sub-optimal) ICA. This report is shown in Table 10.3 in Chapter 10 and is not repeated here for brevity.

Cruise Report

This report is shown in Table 15.8. The cruise flight is done in constant-altitude segments at suitable flight levels, followed by constant-climb rate between flight levels (§ 12.7). In this case, the aircraft operates between FL-330 and FL-370 and does two step climbs. Each constant-altitude cruise segment is approximately 750 n-miles. The climb rate between flight levels is specified.

Table 15.6. *Taxi-out report of fuel/weight-planning analysis*

Parameter	Value	Unit
Fuel rest-to-roll speed	14.52	[kg]
Fuel in idle mode	121.60	[kg]
Fuel in roll mode	368.02	[kg]
Total taxi fuel	504.14	[kg]
Time in idle	6.7	[min]
Time in roll	3.3	[min]
Roll speed	5.0	[m/s]
Roll distance from ramp	3.0	[km]
Taxi procedure	AEO	

Table 15.7. *Take-off report of fuel/weight-planning analysis*

Parameter	Value	Unit	Notes
Gross take-off weight	243.658	[ton]	
Airfield altitude	50.00	[m]	
Air temperature	0.00	± ISA	
Wind speed	–2.00	[m/s]	–3.9 [kt]
Runway state	dry		
Thrust angle	0.000	[degs]	
Stall margin	1.150		
Max lift coefficient	2.435		
Stall speed	69.64	[m/s]	135.3 [kt]
x_{CG}	35	[% MAC]	
Pitch moment of inertia	104.70	[10^6 kgm^2]	
Flap setting, δ_f	20	[deg]	
Rotation velocity, V_R	63.46	[m/s]	123.3 [kt]
Rotation distance, x_R	822.56	[m]	
Rotation time, t_R	29.00	[s]	
Lift-off velocity, V_{LO}	71.46	[m/s]	138.8 [kt]
Lift-off distance, x_{LO}	1,038.88	[m]	
Lift-off time, t_{LO}	32.85	[s]	
Velocity over screen	71.64	[m/s]	139.2 [kt]
Mach over screen	0.211		
Distance to screen, x_{TO}	1,150.18	[m]	
Time over screen, t_{TO}	34.65	[s]	
Climb angle over screen, γ_{TO}	12.48	[deg]	
Fuel burn	214.60	[kg]	
VMCG	55.88	[m/s]	108.6 [kt]
Max (main) tyre temperature	293.2	[K]	+0.6 [K]
Max (nose) tyre temperature	373.7	[K]	+26.3 [K]
Main tyre temp. over screen	292.6	[K]	
Nose tyre temp. over screen	347.4	[K]	
Brake-release net thrust	877.873	[kN]	
Brake-release fuel flow	7.947	[kg/s]	

Table 15.8. *Cruise report of fuel/weight-planning analysis;* $X_{req} = 2317.7$ *[n-miles]; ICW = 239.096 [ton], M = 0.803*

h	FL	X [n-m]	t [min]	m_f [kg]	$\dot{m}_f$ [kg/s]	v_c [ft/min]
10,058	330	772.57	99.26	12,837.1	2.156	
		19.68	2.54	238.1	1.562	787
10,668	350	772.57	99.73	12,435.0	2.078	
		19.52	2.54	230.3	1.510	787
11,277	370	733.37	95.58	11,674.9	2.036	
Totals		2,317.72	299.65	37,415.5	2.081	

Descent Report

En-route descent is performed according to the principles described in Chapter 11. From the final cruise altitude (FCA), the airplane follows some steps down to a reference altitude, for example 1,500 feet (457 m) above the airfield, with a gradual decrease in air speed. A typical descent report is shown in Table 11.2 in Chapter 11; this is not repeated here for brevity.

Analysis of Contingency Fuel

There are various reserve fuel policies, some of which are discussed in § 15.5. A combination of various contingency policies as a function of the required range is shown in graphical form in Figure 15.21. The 20-minute extension of the flight is assumed to take place at the cruise Mach and the final cruise altitude. This fuel reserve is not dependent on the required range because the all-up mass at the start of the flight extension is virtually a constant parameter. The holding cases

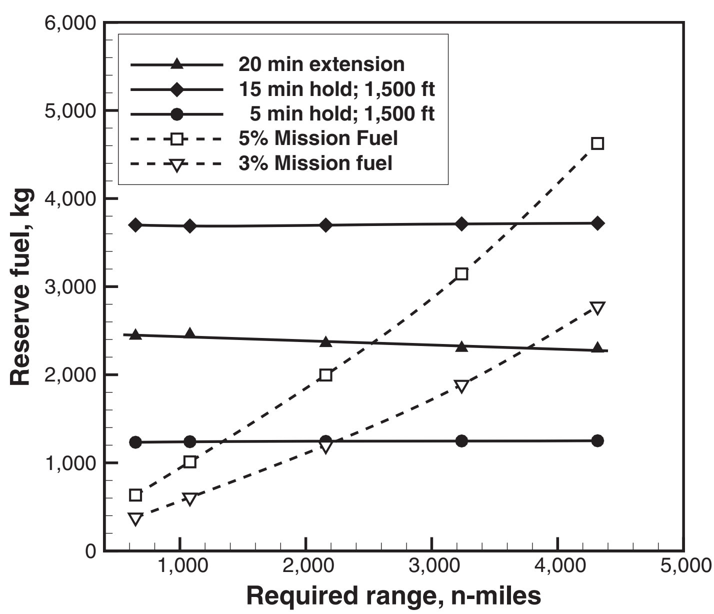

Figure 15.21. Effect of cruise range on contingency fuel; standard day, no wind; full passenger load, 300 kg of bulk cargo (calculated).

Table 15.9. *Basic performance data of model float-plane*

Data	Value	Unit
Cruise TAS	163.4	m/s (592 km/h, 319.5 kt)
Cruise altitude	6,000	m (19,685 feet)
Specific fuel consumption	0.3567	kg/h/kW
Fuel capacity	27,200	kg

(15 or 5 minutes) are performed at 1,500 feet above the airfield at the hold $M =$ 0.350. Other flight conditions are specified in the caption. For the holding cases, we assumed that the initial mass corresponds to the all-up mass before the final approach, which yields a nearly constant reserve fuel.

If an alternate en-route airport is available, the best option is a 3% reserve fuel below 2,200 n-miles. If this is not possible, a 5% reserve fuel is less restrictive than the 20-minute cruise extension for stage lengths below 2,600 n-miles. This conclusion is quite general: each aircraft has a critical stage length beyond which a change in contingency policy may yield a weight advantage. This stage length depends on the actual conditions, in particular weather patterns and payload.

15.11 Case Study: Payload-Range Analysis of Float-Plane

For this study we consider the Lockheed C130J with floats. This is one of the many derivatives of the Hercules aircraft family, designed to perform special operations such as troop transport, anti-submarine warfare, mine counter-measures, search and rescue, fire-fighting, inter-island transport, oil spill response and more. Some basic performance data for the calculation of the floats' drag are given in Table 15.9. The estimated floats' dimensions are summarised in Table 15.10. These data have been calculated from three-view drawings of the airplane, which can be found in the public domain. The wetted area and the volume are more difficult to calculate because these floats are not bodies of revolution.

15.11.1 Estimation of Floats Drag from Payload-Range Chart

One useful application of the payload-range chart is the estimation of the drag difference between different configurations of an airplane. Figure 15.22 shows the estimated payload-range performance of the aircraft with and without floats. The difference in ferry range (zero payload) is about 570 n-miles. This difference is a reflection of a considerable drag created by the floats, the bracing and the increased structural weight.

Table 15.10. *Estimated floats' dimensions*

View	A [m^2]	x [m]	y [m]	z [m]
Top	49.530	21.20	3.39	
Front	4.584		2.97	2.00
Side	26.384	20.55		1.89

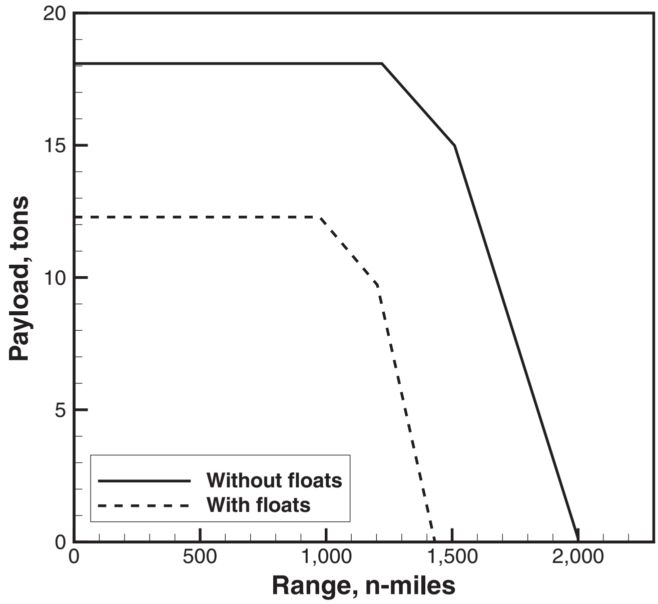

Figure 15.22. Payload range chart of Lockheed C130J, with and without floats.

A first-order estimate of the drag increase is done by assuming that 1) the structural weight of the aircraft is unchanged; 2) the cruise speed is unchanged; 3) the cruise altitude is unchanged; and 4) the drag caused by the bracing is negligible. These effects will be re-evaluated at a later time.

Assume that the nominal performance (C130J without floats) is indicated with the subscript "1", whilst the new configuration (with floats) is indicated with subscript "2". The work done to fly the airplane over the distance x_1 is $\mathcal{W}_1 = Tx_1$. The difference in work done to fly the airplane from x_1 to x_2 is

$$\Delta\mathcal{W} = \Delta T \Delta x = (T_1 - T_2)(x_1 - x_2). \tag{15.55}$$

In steady level flight $\Delta T = \Delta D$. The power relationship for a propeller aircraft is

$$T = \frac{\eta}{U} P, \tag{15.56}$$

where T is the net propulsive thrust, η is the propulsive efficiency, U is the true air speed and P is the shaft power. The relationship between shaft power and fuel flow is

$$\dot{m}_f = \text{SFC}\, P. \tag{15.57}$$

At constant TAS, constant propeller efficiency and constant SFC, we have

$$\Delta T = \Delta D = \frac{\eta}{U}\Delta P = \frac{\eta}{U}\frac{\Delta\dot{m}_f}{\text{SFC}}. \tag{15.58}$$

The difference in fuel flow required is

$$\Delta\dot{m}_f = \dot{m}_{f_2} - \dot{m}_{f_1}. \tag{15.59}$$

The total mass of fuel use by the aircraft is about the same (if we exclude the effects on climb-descent performance). Thus,

$$\dot{m}_{f_1} t_1 \simeq \dot{m}_{f_2} t_2 = m_f. \tag{15.60}$$

The condition of constant TAS is quite useful at this point because $x = Ut$.

$$\dot{m}_{f_2} = \dot{m}_{f_1} \left(\frac{t_1}{t_2}\right) = \dot{m}_{f_1} \left(\frac{x_1}{x_2}\right). \tag{15.61}$$

Thus, the difference in fuel flow is

$$\Delta \dot{m}_f = \dot{m}_{f_1} \left(\frac{x_1}{x_2}\right) - \dot{m}_{f_1} = -\dot{m}_{f_1} \left(\frac{\Delta x}{x_2}\right). \tag{15.62}$$

If we introduce Equation 15.62 into Equation 15.58, we find

$$\Delta D = \frac{\eta}{U} \frac{\dot{m}_{f_1}}{\mathrm{SFC}} \left(\frac{\Delta x}{x_2}\right). \tag{15.63}$$

Note that $\Delta x < 0$ and, therefore, the change in drag is positive. Equation 15.63 is valid only for constant speed, constant cruise altitude and constant SFC. Calculation of the drag increase would require the value of Δx. In first analysis, this value can be taken directly from the payload-range chart (for the ferry condition). However, we must also consider that the aircraft must take off, climb to cruise altitude, descend and land with a minimum of fuel reserves. Therefore, not all of the fuel capacity can be used for cruise. We assume that the fuel capacity is cleared for the mandatory reserves (say 5% of the usable fuel for a ferry operation); an additional 13% is needed for climb to altitude, descend and land. Thus, the usable fuel is about 82% of the fuel capacity.

The cruise range is lower than the mission range because the airplane performs an en-route climb and descent. In first analysis the discount is about 100 n-miles. Using the basic performance data shown in Table 15.9, we estimate that

$$m_{f_1} \simeq 1.042\,\mathrm{kg/s}, \qquad \Delta D \simeq 24{,}840\,\mathrm{kN}, \qquad \Delta C_D \simeq 0.00174. \tag{15.64}$$

In conclusion, the analysis indicates that the floats add about 174 drag counts to the cruise-drag coefficient. This is a considerable amount of drag: it corresponds to a decrease in ferry cruise range by 570 n-miles.

A more accurate calculation is possible, starting from the result just achieved. The first step forward is to consider the optimal air speed of the aircraft with floats. The optimal speed (minimum-power speed) is

$$U_{mp} = \left(\frac{2W}{\rho A} \frac{1}{C_{L_{mp}}}\right)^{1/2}, \tag{15.65}$$

with

$$C_{L_{mp}} = \left(\frac{3C_{D_o}}{k}\right)^{1/2}. \tag{15.66}$$

The difference in profile drag is $\Delta C_D = \Delta C_{D_o}$. The aircraft with the floats must fly faster, as shown by the equation

$$\frac{U_{mp2}}{U_{mp1}} = \left(\frac{C_{L_{mp2}}}{C_{L_{mp1}}}\right)^{1/2} = \left(\frac{C_{D_o} + \Delta C_{D_o}}{C_{D_o}}\right)^{1/2} = \left(1 + \frac{\Delta C_{D_o}}{C_{D_o}}\right)^{1/2} > 1. \tag{15.67}$$

For a 20% increase in profile drag, the optimal TAS would have to increase by about 10%. Thus, if we take into account these new flight conditions, the new fuel flow is

$$\dot{m}_{f_2} = \dot{m}_{f_1}\left(\frac{U_1 x_1}{U_2 x_2}\right) = \dot{m}_{f_1}\left(\frac{x_1}{x_2}\right)\left(1+\frac{\Delta C_{D_o}}{C_{D_o}}\right)^{-1/2}, \tag{15.68}$$

$$\Delta\dot{m}_f = \dot{m}_{f_2} - \dot{m}_{f_1} = \dot{m}_{f_1}\left[\left(\frac{x_1}{x_2}\right)\left(1+\frac{\Delta C_{D_o}}{C_{D_o}}\right)^{-1/2} - 1\right]. \tag{15.69}$$

Equation 15.69 can be used to re-iterate and calculate a new value for the increase in profile drag.

15.12 Risk Analysis in Aircraft Performance

Virtually all of the problems in aircraft performance presented so far are based on deterministic analysis. In other words, given a set of initial conditions and airplane state, it is possible to evaluate almost exactly how the airplane will respond. This is far from true. There are two issues at stake: 1) the risk of failure and the corresponding consequences; and 2) the uncertainty on some state parameters. An example of statistical performance analysis is available in Ref.[17].

As in any other engineering field, there is a risk associated to the operation of an airplane. Safety in air transportation is paramount but risk-free operation, like any other pursuit in life, is virtually impossible. It is perhaps an unfortunate fact that a considerable proportion of the technological progress achieved in aerospace engineering is attributed to accidents of various levels of severity. We presented some examples, but the literature on the subject is vast.

The meaning of risk itself must be clearly defined, particularly the relationship between the risk itself and its severity (or its consequences). Consider the case of ETOPS, which allows twin-engine airplanes to fly over land for a prescribed amount of time (60, 120 or 180 minutes) with one engine failure. The risk analysis associated to this type of operations has allowed twin-engine airplanes to operate across North Atlantic routes. ETOPS safety criteria correspond to less than $3 \cdot 10^{-9}$ fatalities per flight hour arising from a total loss in engine thrust from independent causes. The event of one engine failure on a twin engine is more severe than an engine failure on a three- or four-engine airplane[†].

The analysis shall consider at least the "unit of risk", which is the probability of occurrence of a certain event (incident, engine failure, flight diversion, accident) per flight hour or other relevant measure, for example number of cycles[18]. Most current regulations have been derived from the works of the ICAO and adapted by national aviation regulators, with some local variations. Figure 15.23 shows the notional relationship between the unit of risk and the severity of failure. This severity is mapped onto a failure consequence. The shaded area corresponds to an unacceptable level of risk. The ETOPS risk level is shown for reference.

[†] For operational issues, see Airbus: *Getting to Grips with ETOPS*, *Flight Operations Support and Line Assistance*, Issue V, Oct. 1998, Blagnac, France.

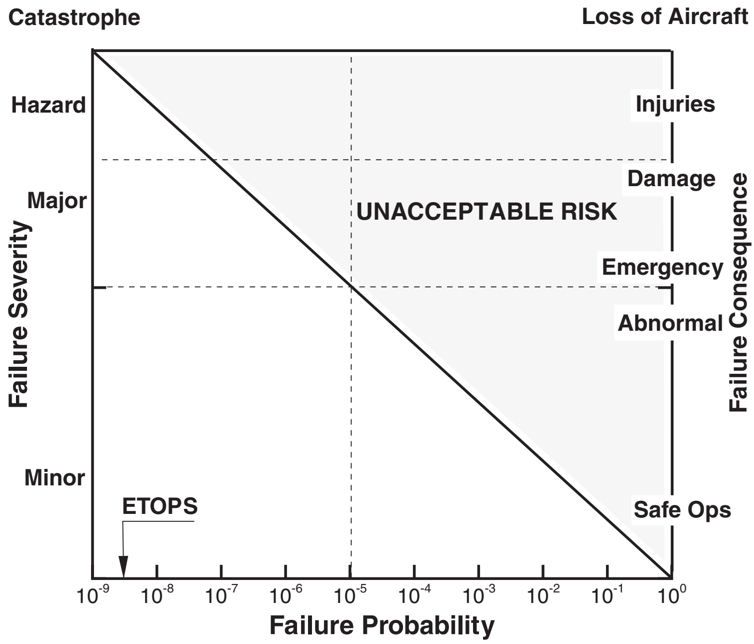

Figure 15.23. Risk analysis in aircraft performance.

Aside from the unit of risk, there is the problem of uncertain parameters (due to stochastic events) and/or imprecise evaluation of parameters. Performance prediction, then, can only be probabilistic. The performance charts produced by the manufacturers (as part of the FCOM) are a case in point. Although the graphs associated to the FCOM show clean lines, the predicted performance must be interpreted in a statistical sense with a high level of confidence.

Some of the performance calculations that are most subject to parameter uncertainty, stochastic external factors and risks of mechanical failure include take-off and landing[19;20]. ESDU has a number of data items on this subject, covering basic statistical analysis and practical methods for evaluating the performance risk[21;22].

Within the context of this discussion, we aim to carry out a "sensitivity" analysis for selected parameters. Because the number of system parameters is quite large, the investigation space is inconveniently large. Therefore, it is necessary in the first instance to isolate the dominant parameters, to evaluate their standard deviation and to establish the effects of such parameters. We give a few examples here.

SPECIFIC AIR RANGE GUARANTEE. The guarantee usually carries a penalty if the performance cannot be matched to the specifications. An example of variability was shown in § 12.8, where we examined the effects of some parameters on cruise performance deterioration. In this instance, we simply assume that a number of tests are carried out; the tests indicate some variability in the SAR; the result is given in the following array.

$$\Delta\text{SAR}, \% = \{1.21, 1.35, -0.59, 0.23, -2.01, 1.05, 1.45, -1.27, -0.97, 0.45\}. \tag{15.70}$$

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Equation 15.70 is a 10-point array containing the percentage change in SAR calculated with 10 flight tests. The statistical properties of this array are: sample standard deviation = 1.21%. On a normal distribution, the mean value has a confidence level of 68.2% within one standard deviation (±1.21%), 95.4% confidence within two standard deviations (±2.42%). This level of confidence will have to be considered carefully before signing any delivery contract.

TAKE-OFF FIELD LENGTH. A preliminary analysis was carried out in Chapter 9, when we considered the effects of an engine failure at a critical point. We then defined the balanced field length and determined the actions to be taken to either stop the airplane or continue the take-off run. Take-off charts contain a number of parametric events that take into account brake-release gross weight, atmospheric temperature, longitudinal wind speed, airfield altitude, runway gradient and flap setting. This type of analysis is in fact deterministic, in the sense that a parameter variation is assumed and the response of the aircraft is measured or simulated. The actual statistical analysis is based on "risk", that is, the probability that certain combinations of winds, airfield altitude, engine failure and so on can conspire (independently or otherwise) to cause an unacceptable level of risk, as indicated in Figure 15.23. Additional considerations may be necessary if the airplane has changed configuration and its condition has deteriorated since the last inspection. The exercise of bringing all these effects together is a major project.

NOISE PERFORMANCE GUARANTEE. A detailed analysis of aircraft noise is carried out in Chapter 16. Noise measurements are also based on statistical values*. The effects of various sources on overall noise can be assessed by calculating the noise response to a set of random variations in parameter settings. This analysis is strongly non-linear because of the log-sum between components and because the noise metric is an integral over the full trajectory.

Summary

We addressed several problems in aircraft-mission analysis, with focus on transport operations. We first examined issues of flight scheduling to maximise the use of the aircraft, and demonstrated that departure and arrival are time-critical. Next, we illustrated the importance of the payload-range chart, which is one of the main drivers in aircraft utilisation. We presented several numerical methods for the computation of payload-range charts and for mission planning; in particular, we demonstrated how the predictor-corrector scheme applied to the mission fuel is the most robust method for calculating the mission fuel. We presented a detailed case study of transport-mission analysis and discussed each flight segment separately. A number of other transport of practical interest include flight with en-route stop and fuel tankering. We highlighted the importance of the corresponding choices, which respond to market conditions. The DOC depend on the fuel consumption but also on a considerable number of parameters in, around and external to the aircraft.

* An example is available in the certification of the Airbus A318, A319, A320, A321. See EASA, Type Certificate for Aircraft Noise, TCDSN EASA.A.064, Issue 10, June 2011.

Calculation of the costs is made difficult by the lack of data that can be generalised. We showed a case study of a float-plane mission and the influence of the floats' drag on its payload range chart. We concluded the chapter by illustrating the importance of risk analysis in aircraft performance.

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Nomenclature for Chapter 15

A = wing area
c = irretrievable loss of engine performance; constant
c_{usf} = usable fuel ratio, Equation 15.16
C_1 = ownership cost
C_2 = interest rate
C_3 = fuel cost
C_4 = crew cost
C_5 = spare-parts cost
C_6 = maintenance cost
C_7 = landing fees
C_{56} = combined spare-parts and man-power cost, Equation 15.54
C_D = drag coefficient
C_{D_o} = profile-drag coefficient
C_{fuel} = fuel cost per kg
$C_{L_{mp}}$ = lift coefficient for minimum power
D = aerodynamic drag
E = fees due to environmental emissions; residual in fuel update
f_j = thrust-specific fuel consumption
F = financing
g = acceleration of gravity
i = iteration counter
I = interest rate on loan
k = induced-drag coefficient
k_w = counter in DOC analysis (§ 15.8)
L = aerodynamic lift
$\mathcal{L}$ = labour rate (currency units per hour)
r = fuel reserve, in percent
m = mass
m_f = fuel burn
m_{fr} = fuel-mass ratio, Equation 15.38
$\dot{m}_f$ = fuel flow
M = Mach number
MH = man-hours
n = lifetime of loan
n = number of iterations/bisection steps
n_w = total number of engine washes
n_1,n_2 = number of pilots/flight attendants, Equation 15.50
N = number of cycles for engine wash
p_a, p_d = fuel price (arrival, departure)
$\mathcal{P}$ = price
P = power
P_{ET} = point of equal time

P_{NT} = point-of-no-return
r = fuel-reserve ratio
R_{div} = diversion distance
R_n = residual value after n years
S = salary (wages) based on full-time rates
SP = spare parts
t = time
T = net thrust
T_b = utilisation time (block time plus turnaround time)
T_o = full-time working hours per year
U = true air speed (also TAS)
U_{mp} = speed of minimum power
$\mathcal{W}$ = mechanical work
W = weight
W_d = weight at the top of descent
W_e = empty weight
W_f = fuel weight
W_{land} = landing weight
W_{mfw} = maximum fuel weight
W_{msp} = maximum structural payload weight
W_{mlw} = maximum landing weight
W_p = payload weight
W_{to} = take-off weight
x = number of cycles in thousands, $N/1{,}000$; distance
x_c = en-route climb distance
x_d = en-route descent distance
x_i = flight-distance segments (climb, cruise, descent ...)
X_{design} = manufacturer's design range
X = stage length
X_{div} = diversion distance
X_{hold} = holding distance
X_{out} = equivalent all-out range
X_p = payload range
X_{req} = required range

Greek Symbols

α_1 = deterioration, Equation 15.49
α_2 = fuel-efficiency-deterioration-lapse coefficient
η = propulsive efficiency
η_M = log-derivative of the propulsive efficiency
ϵ = fraction of the estimated mission fuel
μ_r = rolling resistance
ξ = fuel fraction, Equation 15.4
ρ = air density
σ = relative air density

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Subscripts/Superscripts

$[.]_c$	=	climb
$[.]_d$	=	descent
$[.]_e$	=	empty
$[.]_{ET}$	=	equal-time
$[.]_{ext}$	=	extended range
$[.]_f$	=	fuel
$[.]_k$	=	index for maintenance items or spare parts
$[.]_i$	=	iteration counter
$[.]_{mlw}$	=	maximum landing weight
$[.]_{NT}$	=	no return time
$[.]_{msp}$	=	maximum structural payload
$[.]_p$	=	payload; predicted value
$[.]_{div}$	=	diversion
$[.]_{hold}$	=	hold
$[.]_{res}$	=	reserve
$[.]_{to}$	=	take-off
$[.]^*$	=	estimated quantity